Robust supercritical width versus tree-like size tradeoff for graph isomorphism resolution

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Editorial note. Only robust ordinary-width main result counts. Narrow-width precursor and all game/graph supporting results retained with full original proofs and appendices. Article wrapper replaces LIPIcs metadata; native TikZ mathematics unchanged.

Editorial note. Rendering fixes supply claimproof/QED display macros and remove one stray semicolon following a TikZ style declaration; original source retained. MnSymbol font package replaced by amssymb; unused subfiles package omitted.

Allowed standard prerequisites (editorial).

- Author-cited discrepancy lifting and standard resolution background. All source-local constructions and appendix proofs retained.

Abstract

We study the refutation complexity of graph isomorphism in the tree-like resolution calculus. Torán and Wörz [43] showed that there is a resolution refutation of *narrow width* k for two graphs if and only if they can be distinguished in $(k+1)$-variable first-order logic (FO^{k+1}). While DAG-like narrow width k resolution refutations have size at most n^k, tree-like refutations may be much larger. We show that there are graphs of order n, whose isomorphism can be refuted in narrow width k but only in tree-like size $2^{\Omega(n^{k/2})}$. This is a *supercritical* trade-off where bounding one parameter (the narrow width) causes the other parameter (the size) to grow above its worst case. The size lower bound is super-exponential in the formula size and improves a related supercritical trade-off by Razborov [36]. To prove our result, we develop a new variant of the k-pebble EF-game for FO^k to reason about tree-like refutation size in a similar way as the Prover-Delayer games in proof complexity. We analyze this game on the compressed CFI graphs introduced by Grohe, Lichter, Neuen, and Schweitzer [25]. Using a recent improved *robust* compressed CFI construction of de Rezende, Fleming, Janett, Nordström, and Pang [19], we obtain a similar bound for *width* k (instead of the stronger but less common *narrow width*) and make the result more robust.

1 Introduction

A common theme in proof complexity is the difficulty of refuting a given CNF formula in a particular proof system. There are many variants of this problem depending on the proof system and the notion of difficulty under investigation. We focus on (variants of) resolution, perhaps the most-studied proof system. Here typical measures of difficulty include the minimum width, depth, space, and (tree-like) size over all refutations of the input formula.

By analyzing the proof complexity of formulas that encode natural combinatorial problems, we also gain insights about the inherent complexity of these problems. In this paper, we focus on the graph isomorphism problem, the complexity status of which is still unknown [31]. On the one hand, Babai [4] showed in a breakthrough result that graph isomorphism is solvable in quasi-polynomial time (see also [27]), which makes it a rare natural candidate for a problem that might be neither NP-complete nor polynomial-time solvable. On the other hand, only relatively weak complexity lower bounds are known [40]. This motivates the study of the hardness of graph isomorphism from other perspectives, such as proof complexity.

In this direction, Torán [41] showed that graph isomorphism is hard for resolution: There are non-isomorphic graphs on n vertices such that every resolution refutation certifying non-isomorphism has size $2^{\Omega(n)}$. Graph isomorphism and the Weisfeiler-Leman algorithm—a well-known algorithm in the context of graph isomorphism—have both been studied in different proof systems, including Sherali-Adams [3, 29], (extended) polynomial calculus [12, 13, 34], sum-of-squares [33], cutting planes [42], and in extensions of

resolution with different symmetry rules [37,42]. We are interested in 'tree-like' refutations, which intuitively means that whenever we want to use a clause in a proof step we have to re-derive it. Tree-like resolution corresponds exactly to Boolean decision trees and is closely related to the DPLL algorithm. We will be concerned with 'trade-offs' between the *width*—i.e. the maximum number of literals occurring in any clause—and the *size* of tree-like refutations.

Studying trade-offs—i.e. situations where in order for a refutation to be 'easy' with respect to one measure, it has to be 'hard' with respect to another—is a prominent theme in proof complexity. For example, there are trade-offs between size and space [5,9], between width and depth, and between width and size [11,16,39]. There are two senses in which a trade-off can be particularly strong. Firstly, a trade-off is *robust* if it not only shows that one measure A has to be large when another measure B is small, but also that B can be increased over some (hopefully) wide range without that the bound on A decreasing. Secondly, a trade-off is *supercritical* if, in case we restrict B, the measure A must be larger than the general upper bound on A (over all formulas) in the case that B is not restricted.

In fact, one of the first supercritical trade-offs for resolution—proved by Razborov [36]—concerns tree-like size and width: For every $k = k(n)$, there are k-CNF formulas over n variables that can be refuted in width $O(k)$, but such that every tree-like refutation of minimum width requires depth $n^{\Omega(k)}$ and size $2^{n^{\Omega(k)}}$. When the width is unrestricted then, for each unsatisfiable formula over n variables, there is a tree-like refutation of size at most 2^n. Hence, when bounding the width, the tree-like refutation size increases beyond its worst-case and gets super-exponential in the variable number. This trade-off is not only supercritical but also robust: The lower bounds also hold for width-k' refutations for larger $k' < n^{1-\varepsilon}/k$.

One drawback of Razborov's trade-off is that it is not supercritical if we measure size and width with respect to the formula size rather than the number of variables. While the k-CNF formula of Razborov uses n variables, it has size about $N := n^{\Theta(k)}$. Thus in terms of the formula size N, the bound on tree-like resolution size is roughly 2^N. Our first main result concerns *narrow resolution* [21] which extends resolution by an additional rule avoiding side effects caused by large width clauses in the input formula. It is closely related to first-order logic: The width and the depth of *narrow* graph isomorphism refutations correspond to the number of variables and the quantifier depth respectively of first-order formulas distinguishing graphs [42]. We show a supercritical trade-off with respect to formula size between width and size for tree-like narrow resolution applied to graph isomorphism formulas.

Theorem 1. *For all integers $k \geq 3$ and $n \in \mathbb{N}$, there are two non-isomorphic colored graphs $\mathcal{G}$ and $\mathcal{H}$ of order $\Theta(n)$ and color class size 16 such that*

1. *there is a width-k narrow resolution refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$, and*
2. *every width-k tree-like narrow resolution refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ has size $2^{\Omega(n^{k/2})}$.*

Narrow resolution is stronger than resolution; in particular, the minimal width of a narrow resolution refutation is at most the minimal width of a (plain) resolution refutation. The formula $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$, which encodes graph isomorphism of $\mathcal{G}$ and $\mathcal{H}$, has size $O(n^4)$ for graphs of order n. Thus, Theorem 1 indeed yields a supercritical trade-off between width and tree-like size for narrow resolution *with respect to the formula size*. Building upon very recent work due to de Rezende, Fleming, Janett, Nordström, and Pang [19], who provided a not only supercritical but also robust variant of the trade-off in [25], we can show that Theorem 1 also applies to usual (non-narrow) resolution and the trade-off can be made somewhat robust.

Theorem 2. *For all integers $k \geq 3$, $1 \leq t \leq \frac{2}{5}k - 1$, and $n \in \mathbb{N}$, there are two colored graphs $\mathcal{G}$ and $\mathcal{H}$ of order $\Theta(n)$ and color class size 16 such that*

1. *there is a width-$(k + 16)$ resolution refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$, and*
2. *every width-$(k + t - 1)$ tree-like resolution refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ has size $2^{\Omega(n^{k/(t+1)})}$.*

With this theorem, we address Razborov's call for supercritical bounds in terms of formula size [36]. Moreover, our trade-off applies to formulas encoding a natural combinatorial problem and is somewhat robust for $t > 16$ and sufficiently large k. Since the maximum size of a width k tree-like refutation is $2^{O(n^k)}$, our lower bounds are almost optimal in this range. Our proof utilizes machinery from *finite model theory*: We introduce a new Ehrenfeucht–Fraïssé style game played on two graphs and show that lower bounds for this game imply lower bounds on the tree-like size of narrow resolution refutations of the corresponding graph isomorphism formula.

Razborov's Trade-Off and Weisfeiler-Leman. To prove his trade-off, Razborov used a compression technique, known as *hardness condensation* [15, 36], that is based on *xorification* and variable reuse and converts large but hard formulas into smaller ones that are still hard. Xorification is a well-known technique which replaces every variable in a formula by an XOR of fresh variables. Xorification, or variable substitution in general, has found many applications in proof complexity (see e.g. [6, 8–10]). Razborov's compression technique was adapted to the Weisfeiler-Leman (WL) algorithm—an important algorithm in the field of graph isomorphism [4, 23, 28]. The algorithm, parameterized by a dimension k, is a graph isomorphism heuristic, that is, whenever it distinguishes two graphs they are not isomorphic but, for every k, it fails to distinguish all non-isomorphic graphs [17]. Of particular interest is the number of iterations needed by the k-dimensional WL-algorithm to distinguish two graphs; this almost corresponds to the quantifier depth needed in $(k+1)$-variable first-order logic with counting to distinguish them. Berkholz and Nordström [15] adapted Razborov's compression technique to construct k-ary relational structures for which the k-dimensional WL-algorithm requires $n^{\Omega(k/\log k)}$ iterations; here the best known upper bound is $O(n^{k-1}/\log n)$ [24]. From the perspective of trade-offs, first-order logic (without a bound on the variables) requires at most quantifier depth n to distinguish all non-isomorphic graphs, which means this trade-off is also supercritical. The lower bound was recently improved to $n^{\Omega(k)}$ [24].

The trade-offs described above have a common drawback: They are supercritical with respect to the number of variables and the number of vertices of the structures, respectively, but not with respect to the formula size or the size of the structure (in terms of the number of tuples in the k-ary relations). The common reason is that hardness condensation turns 3-CNF formulas into k-CNF formulas and 3-ary structures into k-ary ones. But recently, Grohe, Lichter, Neuen, and Schweitzer [25] introduced a powerful new compression technique for the so-called Cai-Fürer-Immerman (CFI) graphs [17] to prove a lower bound of $\Omega(n^{k/2})$ for the iteration number of the k-dimensional WL-algorithm on graphs of order n. The bound not only improves the known ones, it is also a bound on graphs and, as graphs have size $O(n^2)$, the lower bound is supercritical with respect to the structure size. The inspiration for this paper was to see if this new technique yields analogous proof complexity results.

Our Techniques and New Games. Tree-like refutations can be (almost) exponentially larger than their non-tree-like counterparts [7]. The usual tool to prove width-k tree-like size lower bounds is the Prover-Delayer game [35]. Prover maintains a partial assignment to at most k variables. In each round, Prover forgets one variable and asks Delayer for an assignment to another one. Delayer can either give such an assignment or allow Prover to set it; in the latter case Delayer scores a point. Prover wants to find an inconsistent partial assignment and Delayer wants to gain as many points as possible. If Delayer has a strategy to score p points, then every width-k tree-like refutation has size at least 2^p. On xorified formulas, where each variable v is replaced with an XOR of v_0 and v_1, Delayer can always gain a point when Prover queries v_0 or v_1 so long as the other one is not already assigned. This leads to tree-like size lower bounds exponential in the depth of a refutation [44].

However, the xorification of a graph isomorphism formula is not necessarily a graph isomorphism formula. Since we are interested in such formulas, our idea is to instead apply xorification on the level of graphs. We show that twins in a graph, i.e., vertices with the same neighborhood, can play the role of XORs in a formula: When we isomorphically map a pair of twins in one graph to a pair of twins in the other graph, the image of the first twin can be chosen arbitrary. We consider *twinned graphs*, where every vertex is replaced by a pair of twins. Ultimately, we want to show that the narrow tree-like refutation size of a twinned graph is exponential in the refutation depth of the original graph. Unfortunately, there seems to be no generic argument for this. To show that this is indeed the case for the graphs we consider, we introduce a variant of the Prover-Delayer game suited for narrow resolution. Then we use techniques from finite model theory to show lower bounds for this game. For other examples of finite-model-theoretic techniques in proof complexity see, e.g. [1, 11, 22].

We cannot reuse the correspondence between width-$(k-1)$ narrow resolution and k-variable first-order logic [43], or equivalently the k-pebble game [30], because we care about tree-like size, not only about width and depth. The issue is that assigning a variable to one or to zero in graph isomorphism formulas is not symmetric: In terms of isomorphisms, fixing the image of a vertex is usually more restrictive than forbidding a single vertex as the image of another vertex. We introduce a new pebble game, called the *k-pebble game with blocking*, which captures this difference between one and zero assignments. Round lower bounds in the pebble game with blocking imply exponential size lower bounds for tree-like resolution.

Another game is involved in this lower bound. The hardness of uncompressed CFI graphs for k-variable first-order logic is captured by the k-Cops and Robber game [25,38], which forgets about the CFI construction and instead considers the simpler underlying *base graphs*. For compressed CFI graphs, this game was modified to the compressed k-Cops and Robber game [25]. To obtain lower bounds for the k-pebble game with blocking, we have to introduce a blocking mechanism to the compressed k-Cops and Robber game. Via all these games,

we obtain the $2^{\Omega(n^{k/2})}$ narrow width-k tree-like size lower bound in Theorem 1.

From Narrow to Plain Resolution. We lift Theorem 1 to (non-narrow) resolution. Since lower bounds for narrow resolution imply lower bounds for resolution, transferring the lower bounds is trivial. But it is unclear whether the relevant isomorphism formula can be refuted in (non-narrow) width k. By increasing the width by the maximal color class size of these graphs (which is 16), we can simulate the narrow resolution refutation by a plain resolution refutation. But now the lower bound from Theorem 1 does not apply anymore. At this point, the aforementioned result from [19] comes to hand: The compression of the CFI graphs get modified to obtain, for every fixed $t < k$, graphs whose isomorphism formula can be refuted in narrow width k but every narrow width $k + t$ refutation has depth at least $\Omega(n^{k/(t+1)})$. So the lower bound is robust within the range from k to $k + t - 1$. This construction can be seen as an interpolation between the original compression [25] with round lower bound $\Omega(n^{k/2})$ and the linear round lower bound $\Omega(n)$ by Fürer [20]; both appear as special cases for $t = 1$ and $t = k$. Our approach with twinned graphs also applies to the improved construction implying a narrow width-$(k+t)$ tree-like size lower bound of $2^{\Omega(n^{k/(t-1)})}$, but we need to restrict the range of t even further. For k large enough and $t > 16$ we can actually refute isomorphism of the graphs in (non-narrow) width $k + t$ and finally obtain a supercritical trade-off between width and tree-like size for resolution with respect to formula size (Theorem 2).

Further Related Work. How the robust compressed CFI construction [19] yields a supercritical width-depth trade-off for resolution was presented at the Oberwolfach workshop *Proof Complexity and Beyond* [2]. The resulting preprint [19] also contains a trade-off for tree-like resolution. A key difference is that our trade-off applies to graph-isomorphism formulas. Also, different techniques are used. We cannot apply hardness condensation techniques to graph isomorphism formulas but apply a form of xorification on the level of graphs and analyze them using model theoretic techniques. In contrast, the trade-off of [19] is obtained via xorification; the parameters obtained are within a constant factor of one-another.

2 Preliminaries

Graphs. An *(undirected) graph* $\mathcal{G}$ is a tuple (V, E) where V is a finite set of *vertices* and $E \subseteq \binom{V}{2}$ is a set of *edges*. The vertex set of $\mathcal{G}$ is denoted by $V_\mathcal{G}$ and the edge set of $\mathcal{G}$ by $E_\mathcal{G}$. For $W \subseteq V_\mathcal{G}$, the *subgraph of $\mathcal{G}$ induced by W* is denoted by $\mathcal{G}[W]$. The *distance* between two vertices $u, v \in V_\mathcal{G}$ is the number of edges in a shortest path between u and v in $\mathcal{G}$. The distance between $U, W \subseteq V_\mathcal{G}$ is the minimal distance of all $u \in U$ and $v \in W$. We will sometimes consider directed graphs, where the set of edges is a subset of $V_\mathcal{G}^2$, but we mention this explicitly. A directed graph is *acyclic*, if it does not contain a (directed) cycle. A *source* (or *sink*) is a vertex without incoming (or outgoing) edges. A *colored* graph $\mathcal{G}$ is a tuple (V, E, χ) such that (V, E) is a graph and χ is a map $V \to \mathbb{N}$. We interpret χ as a vertex coloring of $\mathcal{G}$ and denote it by $\chi_\mathcal{G}$. The *color class* of $u \in V_\mathcal{G}$ is the set $\chi_\mathcal{G}^{-1}(\chi_\mathcal{G}(u))$ of vertices of the same color as u. The *color class size* of $\mathcal{G}$ is the maximal cardinality of its color classes. The graph $\mathcal{G}$ is *ordered* if χ is injective. We can see every graph as a colored graph in which every vertex is colored 0. An *isomorphism* of colored graphs $\mathcal{G}$ and $\mathcal{H}$ is a bijection $f\colon V_\mathcal{G} \to V_\mathcal{H}$ such that, for all $u, v \in V_\mathcal{G}$, we have $\chi_\mathcal{G}(u) = \chi_\mathcal{H}(f(u))$, and $\{u, v\} \in E_\mathcal{G}$ if and only if $\{f(u), f(v)\} \in E_\mathcal{H}$. If there is such an isomorphism, $\mathcal{G}$ and $\mathcal{H}$ are *isomorphic*.

Resolution. A *literal* is a proposition variable x or its negation $\overline{x} := \neg x$. We set $\overline{\neg x} = x$. A *clause* C is a finite set of literals $\{\lambda_1, \dots, \lambda_k\}$. We may write clauses as disjunctions, e.g., $C = (\lambda_1 \vee \cdots \vee \lambda_k)$. A *CNF formula* F is a finite set of clauses $\{C_1, \dots, C_m\}$, which we may write as a conjunction $F = (C_1 \wedge \cdots \wedge C_m)$. The set of *variables occurring* in a clause C is $\mathsf{var}(C)$ and for a CNF formula F it is $\mathsf{var}(F) := \bigcup_{C \in F} \mathsf{var}(C)$. A *(partial) assignment* for a CNF formula F is a (partial) map $\sigma\colon \mathsf{var}(F) \to \{0, 1\}$. The *domain* of σ is $\mathsf{dom}(\sigma)$. The *size* of σ is $|\mathsf{dom}(\sigma)|$. The assignment σ *violates* a clause $C \in F$ if $\mathsf{var}(C) \subseteq \mathsf{dom}(\sigma)$ and σ satisfies no literal in C. For a variable $x \in \mathsf{var}(F)$ and a Boolean value $\delta \in \{0, 1\}$, let $\sigma[x \mapsto \delta]$ be the assignment with domain $\mathsf{dom}(\sigma) \cup \{x\}$ derived from σ that sets x to δ, i.e., $\sigma[x \mapsto \delta](x) = \delta$ and $\sigma[x \mapsto \delta](y) = \sigma(y)$ for all $y \in \mathsf{dom}(\sigma) \setminus \{x\}$. For partial assignments σ and σ', we write $\sigma' \subseteq \sigma$ if $\mathsf{dom}(\sigma') \subseteq \mathsf{dom}(\sigma)$ and $\sigma'(x) = \sigma(x)$ for all $x \in \mathsf{dom}(\sigma')$. For $k \in \mathbb{N}$, we write $[k] := \{1, \dots, k\}$. We now introduce the proof systems studied in this paper.

Definition 3 (Narrow Resolution [21])**.** *A* narrow resolution derivation π *of a clause D from a CNF formula F is a directed acyclic graph $\pi = (V, E)$ whose vertices are labeled with clauses, D is the label of a source of π, and all sinks of π are labeled with clauses in F. Moreover, for every vertex $v \in V$, its clause C is derived from the clauses $C_1, \dots, C_\ell$ labeling the vertices, to which v has an outgoing edge, by one of the following three rules.*

1. **Axiom Rule:** $\ell = 0$ *and* $C \in F$.
2. **Resolution Rule:** $\ell = 2$ *and* $C_1 = A \vee x$, $C_2 = B \vee \overline{x}$, *and* $C = A \vee B$.
3. **Narrow Resolution Rule:** $\ell \geq 2$ *and, up to reordering,* $C_\ell = (A \vee \lambda_1 \vee \cdots \vee \lambda_{\ell-1}) \in F$ *is an axiom,* $C = (A \vee A_1 \cdots \vee A_{\ell-1})$, *and* $C_i = (A_i \vee \overline{\lambda_i})$ *for all* $i \in [\ell - 1]$.

A *resolution derivation* is a narrow resolution derivation using only Rules 1 and 2. The derivation π is *tree-like* if π is a tree. In this case we call the unique source the *root* and the sinks *leaves*. The *size* $|\pi|$ of the derivation π is the number of vertices $|V|$. The *depth* of π is the length of the longest directed path in it. The *width of a clause* C is its number of literals. The *width of a derivation* $w(\pi)$ is the maximal width of all clauses in π. The *narrow width of a derivation* $w^*(\pi)$ is the maximal number of literals among all those clauses in π that are not axioms. A *k-narrow* derivation is a narrow resolution derivation of narrow width at most k. A derivation of the empty clause from F is a *refutation* of F.

3 A Prover-Delayer Game for Tree-Like Narrow Resolution

In this section, we introduce a game that allows us to prove lower bounds on the size of k-narrow tree-like refutations of *graph isomorphism formulas*. We now introduce these formulas. Let $\mathcal{G}$ and $\mathcal{H}$ be n-vertex colored graphs; following [43], we define a CNF formula $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ whose solutions correspond to isomorphisms $\mathcal{G} \to \mathcal{H}$. For all vertices $u \in V_\mathcal{G}$ and $v \in V_\mathcal{H}$, we add a propositional variable $x_{u,v}$. The variables $x_{u,v}$ have the intended meaning that u is mapped to v. The CNF formula $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ contains three types of clause.

- *Color Clauses:* for each vertex $u \in V_\mathcal{G}$, let $W_u := \chi_\mathcal{H}^{-1}(\chi_\mathcal{G}(u))$ be the vertices of $\mathcal{H}$ with the same color as u. Add the clause $\bigvee_{v \in W_u} x_{u,v}$ to encode that u is mapped to a vertex of the same color. For each $v \in V_\mathcal{H}$, let $W_v := \chi_\mathcal{G}^{-1}(\chi_\mathcal{H}(v))$ and add the clause $\bigvee_{u \in W_v} x_{u,v}$.
- *Bijection Clauses:* For all $u \in V_\mathcal{G}$ and distinct $v, w \in V_\mathcal{H}$, we add the clause $(\neg x_{u,v} \vee \neg x_{u,w})$ to encode that an isomorphism is a function. For all distinct $u, v \in V_\mathcal{G}$ and $w \in V_\mathcal{H}$, we add the clause $(\neg x_{u,w} \vee \neg x_{v,w})$ to encode injectivity of the desired isomorphism.
- *Edge Clauses:* for all $u, u' \in V_\mathcal{G}$ and $v, v' \in V_\mathcal{H}$ with $u \neq u'$ such that $\{u, u'\} \in E_\mathcal{G}$ if and only if $\{v, v'\} \notin E_\mathcal{H}$, we include the clause $\neg x_{u,v} \vee \neg x_{u',v'}$. This encodes the edge relation.

The formula $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ has $O(n^2)$ variables, $O(n^4)$ clauses, width equal to the maximal color class size of $\mathcal{G}$ and $\mathcal{H}$ (unless every vertex gets a unique color; in this case the width is two), and is satisfiable if and only if $\mathcal{G}$ is isomorphic to $\mathcal{H}$.

The k-Narrow Prover-Delayer Game. Let $\mathcal{G}$ and $\mathcal{H}$ be non-isomorphic colored graphs. The *k-narrow Prover-Delayer game* on $\mathcal{G}, \mathcal{H}$ is played by two players, Prover and Delayer, who construct partial assignments for $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ as follows. The game begins with the empty assignment $\sigma_0 = \varnothing$. Let σ_{t-1} be the assignment after the $(t-1)$-th round. In round t, Prover chooses $\sigma \subseteq \sigma_{t-1}$ with $|\mathsf{dom}(\sigma)| \leq k - 1$ and makes one of the following kinds of moves.

1. *Resolution Move:* Prover chooses a variable $x \notin \mathsf{dom}(\sigma)$. Delayer chooses a response.
   (a) *Committal Response:* Delayer responds with $\delta \in \{0, 1\}$ and sets $\sigma_t := \sigma[x \mapsto \delta]$.
   (b) *Point Response:* Delayer gets a point; Prover picks $\delta \in \{0, 1\}$ and sets $\sigma_t := \sigma[x \mapsto \delta]$.
2. *Narrow Move:* Prover chooses a color clause C from $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$. Again Delayer chooses one of two response types.
   (a) *Committal Response:* Delayer chooses some $x \in C \setminus \sigma^{-1}(0)$ and sets $\sigma_t := \sigma[x \mapsto 1]$.
   (b) *Point Response:* Delayer chooses distinct $x, y \in C \setminus \sigma^{-1}(0)$ and gets a point; Prover chooses $z \in \{x, y\}$ and sets $\sigma_t := \sigma[z \mapsto 1]$.

If the assignment σ_t violates a clause of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$, the game ends and *Prover wins*. Otherwise, the game continues in round $t + 1$. *Prover has an r-point strategy* if, no matter how Delayer plays, Prover can always win the game while limiting Delayer to at most r points. If Prover does *not* have an r-point strategy, then *Delayer has an $(r + 1)$-point strategy*. It will be useful to start the k-narrow Prover-Delayer game on $\mathcal{G}, \mathcal{H}$ from assignments $\sigma \neq \emptyset$. In this case, the game starts at $\sigma_0 = \sigma$. By constructing strategies for Delayer, the game can be used to show tree-like size lower bound for resolution refutations of graph isomorphism formulas; the full proof is given in Appendix A.

Lemma 4. *For all $k \geq 1$, colored graphs $\mathcal{G}, \mathcal{H}$, and k-narrow tree-like refutations π of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$, Prover has a $(\lceil \log(|\pi|) \rceil)$-point strategy in the $(k+1)$-narrow Prover-Delayer game on $\mathcal{G}, \mathcal{H}$.*

Proof Sketch. Prover follows π starting at the empty clause at the root. If a resolution rule is applied to a variable x, then Prover makes a resolution move for x. Similarly, Prover follows narrow resolution moves. If Delayer makes a committal response, Prover moves to the corresponding child in π. If Delayer makes a point response, Prover moves to the child with the smallest subtree 'below it', at least halving the size of the subtree at the current position. Prover wins if a leaf is reached, so Delayer can score at most $(\lceil \log(|\pi|) \rceil$ points. □

The k-pebble Game and Narrow Resolution. We next recall the connection between $(k-1)$-narrow resolution refutations and the k-variable fragment of first order logic [43]. For an integer k, we write $\mathscr{L}_k$ for the set of first order formulas using at most k *distinct* variables. We denote the set of $\mathscr{L}_k$-formulas with quantifier depth at most r by $\mathscr{L}_{k,r}$. If two graphs $\mathcal{G}$ and $\mathcal{H}$ satisfy the same sentences of $\mathscr{L}_k$ or $\mathscr{L}_{k,r}$, the graphs are $\mathscr{L}_k$-equivalent or $\mathscr{L}_{k,r}$-equivalent, respectively, and we write $\mathcal{G} \simeq^k \mathcal{H}$ or $\mathcal{G} \simeq^{k,r} \mathcal{H}$, respectively.

These equivalences are characterized by the following game: Let $\mathcal{G}$ and $\mathcal{H}$ be (colored) graphs and $k, r \in \mathbb{N}$. The *r-round k-pebble game* on $\mathcal{G}, \mathcal{H}$ is played by two players, Spoiler and Duplicator. A *position* of the game is a pair (α, β) of partial assignments $\alpha : [k] \to V_\mathcal{G}$ and $\beta : [k] \to V_\mathcal{H}$ such that $\mathsf{dom}(\alpha) = \mathsf{dom}(\beta)$. These maps define positions of up to k pebble pairs on $\mathcal{G}$ and $\mathcal{H}$. Duplicator aims to show that $\mathcal{G}$ and $\mathcal{H}$ are isomorphic; Spoiler tries to show they are not. Initially, no pebbles are placed. Let (α_t, β_t) be the position at the end of round $t < r$. At the beginning of round $t+1$, Spoiler picks one of the graphs, say $\mathcal{G}$, and $i \in [k]$. The i-th pebble pair is picked up and Spoiler places the i-th pebble for $\mathcal{G}$ on some $u \in V_\mathcal{G}$ yielding the map α_{t+1}. Duplicator responds by similarly placing the i-th pebble for $\mathcal{H}$ on a vertex of $\mathcal{H}$ yielding β_{t+1}. If $(\alpha_{t+1}, \beta_{t+1})$ does *not induce a partial isomorphism*, meaning that $\alpha(i) \mapsto \beta(i)$ is not an isomorphism of the induced subgraphs $\mathcal{G}[\{\alpha(i) \mid i \in \mathsf{dom}(\alpha)\}]$ and $\mathcal{H}[\{\beta(i) \mid i \in \mathsf{dom}(\beta)\}]$, then *Spoiler wins*. Otherwise, if $t+1 < r$, the play continues in the next round. If $t+1 = r$, then *Duplicator wins*. A player (Spoiler or Duplicator) has a *winning strategy*, if they can win independently of the moves of the other player.

Theorem 5 ([30, 43])**.** *Let $k, r \in \mathbb{N}$. The following are equivalent:*

1. *$\mathcal{G} \not\simeq^{k,r} \mathcal{H}$, i.e., $\mathcal{G}$ and $\mathcal{H}$ are* not *$\mathscr{L}_{k,r}$-equivalent.*
2. *Spoiler has a winning strategy in the r-round k-pebble game on $\mathcal{G}, \mathcal{H}$.*
3. *There is a $(k-1)$-narrow resolution refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ of depth at most r.*

It will sometimes be convenient to start the game from position $(\alpha_0, \beta_0) \neq (\emptyset, \emptyset)$; nothing else in the rules of the games changes in this case. Similarly, we may sometimes not specify the number of rounds in advance; in this case the game only ends if Spoiler wins.

4 Twinned Graphs and Pebble Games

In this section, we introduce the *twinned graph* construction, which we described on a high-level in the introduction. This will allow us to transfer lower bounds on the k-pebble game with blocking to lower bounds on the k-narrow Prover-Delayer game (and therefore to lower bounds on $(k-1)$-narrow tree like refutation size).

We use a colored graph $\mathcal{G}$ to define a new colored graph as follows. For each vertex $u \in V_\mathcal{G}$, we set $\mathcal{X}_\mathcal{G}(u) := \{u_0, u_1\}$, where u_0 and u_1 are fresh vertices; intuitively these are copies of u. We define the *twinned graph* $\mathcal{X}(\mathcal{G})$ with vertex set $V_{\mathcal{X}(\mathcal{G})} := \bigcup_{u \in V_\mathcal{G}} \mathcal{X}_\mathcal{G}(u)$ and edge set

$$E_{\mathcal{X}(\mathcal{G})} := \big\{ \{x, y\} \;\big|\; x \in \mathcal{X}_\mathcal{G}(u),\, y \in \mathcal{X}_\mathcal{G}(v), \{u, v\} \in E_\mathcal{G} \big\} \cup \big\{ \mathcal{X}_\mathcal{G}(u) \;\big|\; u \in V_\mathcal{G} \big\}.$$

We give u_0 and u_1 the same color in $\mathcal{X}(\mathcal{G})$ as u has in $\mathcal{G}$. For notational convenience, we define $\hat{u}_0 := u_1$ and $\hat{u}_1 := u_0$. Moreover, we define $\mathcal{X}_\mathcal{G}^{-1}(u_i) := u$ for $i \in \{0, 1\}$.

We first show that—under a mild condition—if there is a k-narrow refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$, then there is a k-narrow refutation of $\mathsf{ISO}(\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H}))$. To state the condition, we need the following notion. Two distinct vertices $u, v \in V_\mathcal{G}$ are *twins* if for every $w \in V_\mathcal{G} \setminus \{u, v\}$, we have that $\{u, w\} \in E_\mathcal{G}$ if and only if $\{v, w\} \in E_\mathcal{G}$. That is, the neighborhoods of u and v in $\mathcal{G}$ are, apart from u and v themselves, identical. Twins u and v are *connected twins* if $\{u, v\} \in E_\mathcal{G}$. Note that if $\mathcal{G}$ has no connected twins, then $\mathcal{X}(\mathcal{G})$ has exactly one pair of connected twins for each vertex in $\mathcal{G}$. This leads to the following observation. See Appendix B for a proof.

Lemma 6. *Let $k \geq 3$ and $\mathcal{G}$ and $\mathcal{H}$ be colored graphs that do not have connected twins. If $\mathcal{G} \not\simeq^{k,r} \mathcal{H}$, then $\mathcal{X}(\mathcal{G}) \not\simeq^{k,r+1} \mathcal{X}(\mathcal{H})$.*

By applying Theorem 5 we obtain the following corollary.

Corollary 7. *Let $k \geq 2$ and $\mathcal{G}$ and $\mathcal{H}$ be colored graphs that have no connected twins. If there exists a k-narrow refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ of depth d, then there exists a k-narrow refutation of $\mathsf{ISO}(\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H}))$ of depth $d + 1$.*

It turns out that Prover-Delayer lower bounds on our twinned graphs are implied by round lower bounds for certain pebble games on the original graphs. The normal k-pebble game is the wrong tool for this task; intuitively, the reason is the asymmetry between setting a variable of a graph isomorphism formula to zero or one.

The k-Pebble Game with Blocking. Let $\mathcal{G}$ and $\mathcal{H}$ be colored graphs and $k, r \in \mathbb{N}$. We define the r-round k-pebble game with blocking on $\mathcal{G}$ and $\mathcal{H}$ as follows. The game is played in rounds by Spoiler and Duplicator. A position in the game is a triple (α, β, c) of partial maps $\alpha : [k] \to V_{\mathcal{G}}$, $\beta : [k] \to V_{\mathcal{H}}$, and $c : [k] \to \{\mathsf{regular}, \mathsf{blocking}\}$ with $\mathsf{dom}(\alpha) = \mathsf{dom}(\beta) = \mathsf{dom}(c)$. The first two maps give the positions of the pebbles and c marks each pair of pebbles as either *regular* or *blocking*. Regular pebbles (possibly) define partial isomorphisms as before, but blocking ones forbid certain ones as follows.

Definition 8 (Partial Isomorphism with Blocking)**.** *Let (α, β, c) be a position in the r-round k-pebble game with blocking on $\mathcal{G}$ and $\mathcal{H}$, $R := c^{-1}(\mathsf{regular})$, and $B := c^{-1}(\mathsf{blocking})$. Then (α, β, c)* induces a partial isomorphism with blocking *if $(\alpha|_R, \beta|_R)$ induces a partial isomorphism and if every regular pebble respects every blocking pebble. Formally, this means that for every $p \in B$ and $q \in R$, we have $(\alpha(p), \beta(p)) \neq (\alpha(q), \beta(q))$.*

In the initial position (α_0, β_0, c_0), all maps are empty. Let (α_t, β_t, c_t) be the position after the t-th round. At the beginning of the $(t + 1)$-th round, Spoiler can make either a *regular move* or a *blocking move*. A regular move works in the same way as a move in the k-pebble game; the pebble pair moved in this turn is then marked $\mathsf{regular}$. For a blocking move, Spoiler picks $p \in [k]$ and places the p-th pebble in $\mathcal{G}$ on some vertex $u \in V_{\mathcal{G}}$ and in $\mathcal{H}$ on some vertex $v \in V_{\mathcal{H}}$. Duplicator next decides how to mark this pair. If Duplicator chooses $\mathsf{regular}$, then the round ends. If instead Duplicator chooses $\mathsf{blocking}$, then the round continues and Spoiler can again choose to make either a regular or a blocking move. If $(\alpha_{t+1}, \beta_{t+1}, c_{t+1})$ does *not* induce a partial isomorphism with blocking, then *Spoiler wins* and the game ends. Otherwise if $t + 1 < r$, the game continues in round $t + 2$. If $t + 1 = r$, then *Duplicator wins*.

We write $\mathcal{G} \simeq_{\mathcal{B}}^{k,r} \mathcal{H}$ if Duplicator has a winning strategy in the r-round k-pebble game with blocking. If $\mathcal{G} \simeq_{\mathcal{B}}^{k,r} \mathcal{H}$ for all $r \in \mathbb{N}$, then we write $\mathcal{G} \simeq_{\mathcal{B}}^{k} \mathcal{H}$. As for the k-pebble game, it will also be convenient to consider variants of the k-pebble game with blocking where we start from arbitrary positions or do not specify the number of rounds in advance. Note that while in the (non-blocking) k-pebble game it never makes sense for Spoiler to place a pebble on an already pebbled vertex, this is not the case in the k-pebble game with blocking.

Spoiler and Duplicator meet Prover and Delayer. We end the section by connecting the k-pebble game with blocking to the k-narrow Prover-Delayer game via the following lemma.

Lemma 9. *Let $\mathcal{G}$ and $\mathcal{H}$ be colored graphs and $k \geq 2$ an integer. If $\mathcal{G} \simeq_{\mathcal{B}}^{k,r} \mathcal{H}$, then Delayer has an r-point strategy in the k-narrow Prover-Delayer game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$.*

Proof Sketch. In the k-narrow Prover-Delayer game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$, Delayer simulates positions of the k-pebble game with blocking on $\mathcal{G}$, $\mathcal{H}$. Intuitively, whenever a $\mathsf{regular}$ pebble pair is placed on vertices u and v (and there is not already a pebble pair on u and v), Delayer should score a point since it 'does not matter' whether we map u_0 to v_0 or to v_1. As the round counter of the k-pebble game with blocking only advances when a pebble pair is marked as $\mathsf{regular}$, filling in the details is relatively straightforward. See Appendix B. □

Lemmas 4 and 9 finally connect the pebble game with blocking to tree-like refutation size.

Theorem 10. *Let $k \geq 1$, and $r \in \mathbb{N}$ and $\mathcal{G}$ and $\mathcal{H}$ be colored graphs. If $\mathcal{G} \simeq_{\mathcal{B}}^{k+1,r} \mathcal{H}$, then every k-narrow tree-like refutation of $\mathsf{ISO}(\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H}))$ has size at least 2^r.*

5 Compressing CFI Graphs

By what we have seen so far (Corollary 7 and Theorem 10), to prove Theorem 1 it suffices to show that Duplicator can survive a large number of rounds in the k-pebble game with blocking on suitably chosen colored graphs $\mathcal{G}, \mathcal{H}$. In this section, we describe a framework which allows us to construct such graphs.

Concretely, we recall a recent approach to construct pairs of graphs that require quantifier depth $\Omega(n^{k/2})$ to be distinguished in k-variable first order logic $\mathscr{L}_k$ (and also with counting) [25]. The key idea is a novel compression technique of the so-called Cai-Fürer-Immerman (CFI) graphs [17] and a concrete compression construction for CFI graphs over grids. Having introduced this construction, we give a method for proving lower bounds for the k-pebble game with blocking on compressed CFI graphs. To do this, we first recall a variant of the Cops and Robber game, which can be used to derive lower bounds on the k-pebble game on compressed CFI graphs, and then extend this game with an appropriate notion of blocking.

CFI Graphs. Let $\mathcal{G} = (V_\mathcal{G}, E_\mathcal{G})$ be a connected ordered graph, called a *base graph*, and $f\colon E_\mathcal{G} \to \mathbb{F}_2$ be a function, where $\mathbb{F}_2$ is the two-element field. From $\mathcal{G}$ and f we derive the colored *CFI graph* $\mathsf{CFI}(\mathcal{G}, f)$: Vertices of $\mathcal{G}$ are called *base vertices*. Every base vertex of $\mathcal{G}$ is replaced by a *CFI gadget* and gadgets of adjacent vertices are connected. The vertices of the CFI gadget for a degree d base vertex $u \in V_\mathcal{G}$ are the pairs $(u, \bar{a})$ for all d-tuples $\bar{a} = (a_1, \dots, a_d) \in \mathbb{F}_2^d$ with $\sum_{i=1}^{d} a_i = 0$. The vertex $(u, \bar{a})$ has *origin* u. Vertices inherit the color of their origin. Since every vertex of the base graph has a unique color, the vertices of each gadget form a color class of the CFI graph. For all adjacent base vertices $u, v \in V_\mathcal{G}$, we add the following edges between the gadgets for u and v: Let u be the i-th neighbor of v and v be the j-th neighbor of u according to the order on $V_\mathcal{G}$. There is an edge between vertices $(u, \bar{a})$ and $(v, \bar{b})$ if and only if $a_i + b_j = f(\{u, v\})$, where a_i is the i-th entry of $\bar{a}$ and b_j is the j-th entry of $\bar{b}$. See Figure 1 for an example. We say that two functions $f, g : E_\mathcal{G} \to \mathbb{F}_2$ *twist an edge* $e \in E_\mathcal{G}$ or the edge $e \in E_\mathcal{G}$ is *twisted by* f *and* g if $f(e) \neq g(e)$. It is well-known that $\mathsf{CFI}(\mathcal{G}, f) \ncong \mathsf{CFI}(\mathcal{G}, g)$ if and only if f and g twist an odd number of edges [17].

Compressing CFI Graphs. CFI graphs are a well-studied tool to derive lower bounds for k-variable logic with counting or other logics, see e.g. [15, 17, 18, 20, 32]. This construction and its generalizations have also been used to derive proof complexity lower bounds on graph isomorphism in various proof systems [12, 13, 33, 37, 41, 42]. We now discuss the method of *compressing CFI graphs* [25]. The goal is to reduce the size of the resulting graph while essentially preserving the hardness of it. The main idea is to identify the gadgets of certain base vertices. The hardness of the resulting compressed CFI graphs heavily depends on which gadgets get identified and can be analyzed using a variant of the Cops and Robber game. We now present this framework.

Definition 11 (Graph Compression)**.** *An equivalence relation $\equiv$ on $V_\mathcal{G}$ is a* $\mathcal{G}$-compression *if for all $u, u', v, v' \in V_\mathcal{G}$ it satisfies the following two conditions:*

1. *If $u \equiv v$, then u and v are non-adjacent and of the same degree.*
2. *If $\{u, v\}, \{u', v'\} \in E_\mathcal{G}$, $u \equiv u'$, $v \equiv v'$, and u is the i-th neighbor of v (according to the order on $V_\mathcal{G}$), then u' is the i-th neighbor of v'.*

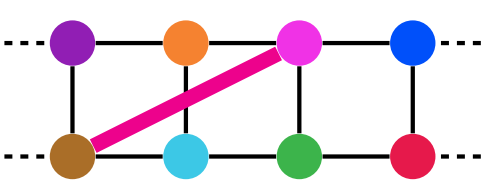
Let $\equiv \subseteq V_\mathcal{G}^2$ be a $\mathcal{G}$-compression. It induces an equivalence relation on $\mathsf{CFI}(\mathcal{G}, f)$ (independently of the function $f\colon E \to \mathbb{F}_2$), which we also denote by $\equiv$, via $(u, \bar{a}) \equiv (v, \bar{b})$ if and only if $u \equiv v$ and $\bar{a} = \bar{b}$. Contracting all $\equiv$-equivalence classes in $\mathsf{CFI}(\mathcal{G}, f)$ into a single vertex yields the colored graph $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$. Formally, the vertices of $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ are the $\equiv$-equivalence classes $u/_\equiv := \{w \in V_{\mathsf{CFI}(\mathcal{G},f)} \mid w \equiv u\}$, and $u/_\equiv$ and $v/_\equiv$ are adjacent if there are $u' \equiv u$ and $v' \equiv v$ such that u' and v' are adjacent in $\mathsf{CFI}(\mathcal{G}, f)$. Observe that $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ is loop-free by the condition on $\equiv$ that equivalent vertices of $\mathcal{G}$ are non-adjacent. The color of a $\equiv$-equivalence class in $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ is the minimal color of one of its members in $\mathsf{CFI}(\mathcal{G}, f)$. To obtain reasonable graphs, f has to be compatible with the compression $\equiv$ in the following sense.

Definition 12 (Compressible)**.** *A function $f\colon E_\mathcal{G} \to \mathbb{F}_2$ is* $\equiv$-compressible *if $f(\{u, v\}) = f(\{u', v'\})$ for all vertices $u, v, u', v' \in V_\mathcal{G}$ such that $\{u, v\}, \{u', v'\} \in E_\mathcal{G}$, $u \equiv u'$, and $v \equiv v'$.*

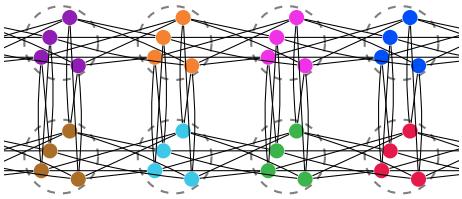
Definition 13 (Compressed CFI)**.** *For a $\mathcal{G}$-compression $\equiv$ and a $\equiv$-compressible function $f\colon E_\mathcal{G} \to \mathbb{F}_2$, the graph $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ expanding the colored graph $\mathsf{CFI}(\mathcal{G}, f)$ with $\equiv$ is a* precompressed CFI graph*, and the colored graph $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ is a* compressed CFI graph*.*

Precompressed CFI graphs can also be seen as edge-colored graphs that use two colors for the edges—one the the regular edges and one for the equivalence relation. An example is shown in Figure 1. The round number of the bijective k-pebble game (a variant of the k-pebble game that characterizes equivalence of k-variable first order logic with counting quantifiers) on precompressed and compressed CFI graphs are almost equal [25]. The corresponding statement for the k-pebble game with blocking is proved similarly. See Appendix C for a proof.

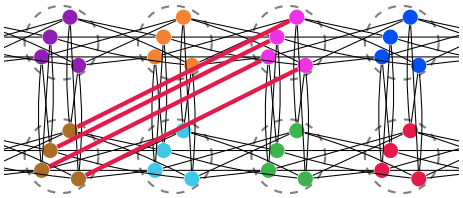
Lemma 14. *Let $k \geq 3$, $r \in \mathbb{N}$, $\equiv$ be a $\mathcal{G}$-compression, and $f, g : E_\mathcal{G} \to \mathbb{F}_2$ be $\equiv$-compressible.*



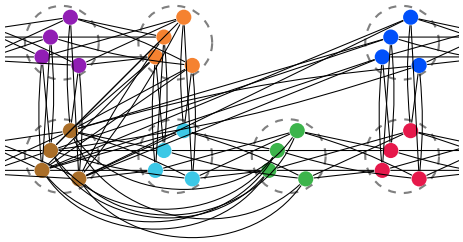
(a) The base graph with the compression.



(b) The corresponding (uncompressed) CFI graph.



(c) The precompressed CFI graph.



(d) The compressed CFI graph.

Figure 1: Compressed CFI graphs for a grid of height 2 as base graph, a very simple compression, which only identifies two base vertices, and the function that assigns 0 to all edges. The compression on the base graph and the induced one on the precompressed CFI graph is drawn in magenta.

1. $\mathsf{CFI}(\mathcal{G}, f) \not\simeq^{k,r}_{\mathcal{B}} \mathsf{CFI}(\mathcal{G}, g)$ *implies* $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \not\simeq^{k,r}_{\mathcal{B}} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$.
2. $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \not\simeq^{k,r}_{\mathcal{B}} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$ *implies* $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv} \not\simeq^{k,r}_{\mathcal{B}} \mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$.
3. $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv} \not\simeq^{k,r}_{\mathcal{B}} \mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$ *implies* $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \not\simeq^{k,r+2}_{\mathcal{B}} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$.

The Compressed Cops and Robber Game. The ability of the bijective k-pebble game to distinguish non-isomorphic CFI graphs is captured by the k-Cops and Robber game [25,38]. A variant of this game—the compressed k-Cops and Robber game—provides lower bounds for compressed CFI graphs. To see this, we consider isomorphisms of CFI graphs. These always twist an even number of edges and can be described in terms of paths in the base graphs by twistings (defined below). Moreover, if these paths are compatible with the compression, they induce isomorphisms of compressed CFI graphs. For ordered base graphs, these twistings correspond one-to-one with isomorphisms of the (compressed) CFI graphs.

Definition 15 (Twisting)**.** *A set $T \subseteq \{(u,v) \mid \{u,v\} \in E\}$ is called a* $\mathcal{G}$-twisting *if, for every $u \in V$, the set $T \cap (\{u\} \times V)$ is of even size. The twisting T*

- twists *an edge $\{u,v\} \in E$ if the set T contains exactly one of (u,v) and (v,u) and*
- fixes *a vertex $u \in V$ if $T \cap (\{u\} \times V) = \emptyset$.*

To obtain a reasonable notion of twistings for isomorphisms of compressed CFI graphs, the twistings have to be compatible with the compression. For more details on (compressed) CFI graphs, their isomorphisms, and twistings, we refer to the original paper [25].

Definition 16 (Compressible Twisting)**.** *For a $\mathcal{G}$-compression $\equiv$, a $\mathcal{G}$-twisting T is called* $\equiv$-compressible *if the following holds for all $u, u' \in V$ with $u \equiv u'$: Let u and u' be of degree d. Then for every $i \in [d]$, we have $(u, v_i) \in T$ if and only if $(u', v'_i) \in T$, where v_i is the i-th neighbor of u and v'_i is the i-th neighbor of u' (according to the order on $\mathcal{G}$).*

The *compressed k-Cops and Robber game* [25] is played on a base graph $\mathcal{G}$ and a $\mathcal{G}$-compression $\equiv$. The Cops Player places cops on up to k $\equiv$-equivalence classes and the robber is placed on one edge of $\mathcal{G}$. Initially, only the robber is placed. The game proceeds in rounds:

1. The Cops Player picks up a cop and announces a new $\equiv$-equivalence class C for this cop.
2. The robber moves. To move from the current edge e_1 to another edge e_2, the robber has to provide a $\equiv$-compressible $\mathcal{G}$-twisting that only twists the edges e_1 and e_2 and that fixes every vertex contained in a cop-occupied $\equiv$-equivalence class.
3. The cop that was picked up in Step 1 is placed on C. The next round starts.

The robber is *caught* if the two endpoints of the robber-occupied edge are contained in cop-occupied $\equiv$-classes. The cops have a winning strategy in r rounds, if they can catch the robber in r rounds independently of the moves of the robber. Similarly, the robber has a *strategy for the first r rounds* if the robber can avoid being caught for r rounds independently of the moves of the Cops Player. The winner of the compressed game depends on the initial position of the robber. This game yields lower bounds for the (bijective) k-pebble game:

Lemma 17 ([25]). *Let $\equiv$ be a $\mathcal{G}$-compression and suppose $f, g : E_{\mathcal{G}} \to \mathbb{F}_2$ only twist a single edge e. If the robber, initially placed on the edge e, has a strategy for the first r rounds in the compressed k-Cops and Robber game on $\mathcal{G}$ and $\equiv$, then $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \simeq^{k,r} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$.*

Introducing Roadblocks for Cops. To obtain lower bounds for the k-pebble game with blocking, we add 'roadblocks' to the compressed Cops and Robber game and prove a blocking analogue of Lemma 17. Let $\mathcal{G}$ be an ordered graph. A *roadblock for a vertex $u \in V_{\mathcal{G}}$* is a nonempty set $N \subseteq \{(u, v) \mid \{u, v\} \in E_{\mathcal{G}}\}$ of (directed) edges incident to u of even size. A $\mathcal{G}$-twisting T *avoids a roadblock* N for a vertex u if $T \cap \{u\} \times V_{\mathcal{G}} \neq N$. In particular, T may contain a strict superset or subset of N. If T does not avoid N, then T *uses* N. A *roadblock for a $\equiv$-equivalence class C* is a nonempty set $N \subseteq [d]$ of even size, where d is the unique degree of the vertices in C. A $\mathcal{G}$-twisting T avoids the roadblock N on C if, for every vertex $u \in C$, the twisting T avoids the roadblock $N_u := \{(u, v_i) \mid i \in N\}$ for u, where v_i denotes the i-th neighbor of u. If T is $\equiv$-compressible and does not avoid N, then T uses N_u for every vertex $u \in C$; we say that T uses N. Let $M \subseteq [d]$ be the set of all $i \in [d]$ such that T contains the edge to the i-th neighbor of some and, since T is $\equiv$-compressible, of every $u \in C$. We write $T(N)$ for the symmetric difference of N and M.

The *compressed and blocking k-Cops and Robber game* is played on a base graph $\mathcal{G}$ and a $\mathcal{G}$-compression $\equiv$. The Cops Player controls cops and roadblocks. The total number of cops and roadblocks is k but the number of each may vary during the game. Cops and roadblocks are placed on $\equiv$-equivalence classes and the robber is located on an edge. Initially, only the robber is placed. A round of the game proceeds as follows: The Cops Player picks up a cop or a roadblock and can choose to play a cop move or a blocking move.

1. A *cop move* proceeds similarly to the non-blocking game. First, the Cops Player announces a $\equiv$-equivalence class C. Next, the robber moves. To move from an edge e_1 to another edge e_2, the robber provides a $\equiv$-compressible $\mathcal{G}$-twisting T that only twists the edges e_1 and e_2, fixes every vertex contained in a cop-occupied $\equiv$-equivalence class, and avoids every roadblock. Afterwards, a cop is placed on the announced class C.

2. For a *blocking move*, the Cops Player announces a $\equiv$-equivalence class C and a roadblock N for C. Next, the robber moves with a $\equiv$-compressible $\mathcal{G}$-twisting T as in the cop move. If T uses N, then a cop is placed on C. Otherwise, the roadblock N is placed on C.

3. The existing roadblocks are updated. If a roadblock N' is placed on a class C', then it is replaced by the roadblock $T(N')$ on C'. Because in both a cop and a blocking move T avoids all roadblocks, $T(N')$ will always be a nonempty set. If a roadblock was placed in this move, the Cops Player can again choose to play either a cop or a blocking move without increasing the round counter.

The notion of the robber being caught or having a strategy for the first r round is the same as in the non-blocking game. As in the non-blocking game, the starting edge of the robber is important. The following lemma is proved similarly to Lemma 17. A full proof is given in Appendix C.

Lemma 18. *Suppose $\equiv$ is a $\mathcal{G}$-compression and $f, g : E_{\mathcal{G}} \to \mathbb{F}_2$ only twist a single edge e. If the robber, initially placed on the edge e, has a strategy for the first r rounds in the compressed and blocking k-Cops and Robber game on $\mathcal{G}$ and $\equiv$, then $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \simeq^{k,r}_{\mathcal{B}} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$.*

The $\equiv$-compressible twistings of the robber induce isomorphisms of the compressed CFI graphs, which respect all currently placed pebbles. These are used to move the twisted edge ('the robber') away from the pebbles. Cops correspond to regular pebble pairs and roadblocks to blocking ones. The case distinction in Point 2 whether a cop roadblock is placed ensures that in a blocking move in the pebble game with blocking the pebble pair gets marked as regular or blocking consistently with the current isomorphism. Updating the roadblocks in Point 3 corresponds to applying the isomorphism induced by the twisting T to them.

6 The Super-Linear Lower Bound with Roadblocks

We now present and analyze the robust compressed CFI construction of [19]. This work shows that the robber can survive for a large number of rounds in the compressed Cops and Robber game for certain compressions.

This section shows that the robber also has a strategy for a large number of rounds in the game with roadblocks. By Lemma 18 and Theorem 10, such a result implies a lower bound on tree-like refutation size for graph isomorphism formulas.

6.1 Compressing Cylindrical Grids

Fix an integer $k \geq 3$ and a sufficiently large integer w. Set $f(k) := 4k$. Let $p_1, \ldots, p_k$ be pairwise coprime numbers such that $\frac{w}{2} \leq p_i \leq w$ for every $i \in [k]$. For all sufficiently large w, such numbers exist [25]. Set $J := f(k) \cdot p_1 \cdot \ldots \cdot p_k$. Let $\mathcal{C}$ be the $k \times J$ cylindrical grid, that is, the $k \times J$ grid, in which we also connect the top and bottom row. The vertices of $\mathcal{C}$ are pairs (i, j) for all $i \in [k]$ and $j \in [J]$. They are ordered lexicographically. We think of the first component as the row index and the second component as the column index. We refer to the first $f(k)$ columns as the *left end* of $\mathcal{C}$, and to the last $f(k)$ columns as the *right end* of $\mathcal{C}$. We use addition on row indices in a modulo-like manner. For example, the $(k+1)$-th row is the first one and $p_{k+1} = p_1$. For each $t \in [k-1]$, we define the following equivalence $\equiv^t$ via:

$$(i,j) \equiv^t (i',j') \quad \iff \quad i = i';\ f(k) < j, j' \leq J - f(k);\ \text{and } j - j' \text{ is divisible by } f(k)p_i \cdot \ldots \cdot p_{i+t}.$$

These equivalences are $\mathcal{C}$-compressions [19]. Note that the vertices in the left or the right end of $\mathcal{C}$ are in singleton $\equiv^t$-equivalence classes. The vertices in between are identified periodically, but the period is different in every row. It is not hard to show that there are $\Theta(w^{t+1})$ $\equiv^t$-equivalence classes. Together with the fact that CFI gadgets for degree 4 base vertices have 8 vertices, this implies the next lemma:

Lemma 19. *For all $t \in [k-1]$ and $\equiv^t$-compressible $f\colon E_{\mathcal{C}} \to \mathbb{F}_2$, the graph $\mathsf{CFI}(\mathcal{C}, f)/_{\equiv^t}$ has order $\Theta(w^{t+1})$ (where k and t are seen as constants) and color class size 8.*

While the order of the graphs is $\Theta(w^{t+1})$, the robber has a strategy for $\Omega(w^k)$ rounds:

Theorem 20 ([19])**.** *For every $t \in [k-1]$, consider the compressed $(k+t)$-Cops and Robber game played on $\mathcal{C}$ and $\equiv^t$. If the robber is initially placed on an edge on the left or right end of $\mathcal{C}$, then the robber has a strategy for the first $\Omega(J) = \Omega(w^k)$ rounds.*

Unfortunately, this theorem does not lift to the game with roadblocks; in order to lift it, we investigate the strategy of the robber in more detail: The robber is always located in either the left or right end of the grid. On uncompressed grids of height k, the optimal strategy of the Cops Player with at most $2k$ cops is to separate the left from the right end of grid using the cops and to move this separator slowly towards the robber (by at most a constant number of columns in each round). So the robber can avoid getting caught for a number of rounds linear in the length of the grid: when a newly announced cop is about to form a separator, the robber moves to the end furthest from the separator. In the compressed game, the strategy is similar. However, the suitable notion of a separator and the analysis of the situations in which the robber can move from the one end of the grid to the other are more complicated. We now describe them on an informal level to illustrate the central ideas. For formal details and more explanations, we refer to Appendix D and to the original works [19, 25].

To move the robber from one end of the grid to the other, we use $\equiv^t$-compressible $\mathcal{C}$-twistings that twist exactly two edges, one in the first and one in the last column. Such twistings are called *t-end-to-end twistings* and are obtained from *ℓ-periodic paths* [25]. Intuitively, these are paths from the first to the last column in the grid $\mathcal{C}$, which repeat every ℓ columns. This means that an ℓ-periodic path is defined by a path in columns 0 to $\ell - 1$ repeating every ℓ columns. If ℓ is the greatest common divisor of the compression periods of all the rows used by the path, then the path induces a t-end-to-end twisting:

Lemma 21 ([19, 25])**.** *Let $t \in [k-1]$, $\pi = (u_1, \ldots, u_m)$ be an ℓ-periodic path, and $I \subseteq [k]$ be the set of all rows of which π contains vertices. If $\ell = \gcd\{f(k)p_i \cdot \ldots \cdot p_{i+t} \mid i \in I\}$, then π induces the $\equiv^t$-compressible $\mathcal{C}$-twisting $\{(u_i, u_{i-1}), (u_i, u_{i+1}) \mid 1 < i < m\}$.*

We now turn to a suitable notion of 'a separator' for the compressed grid. Let W be a set of $\equiv^t$-equivalence classes. A *t-virtual cordon* [19] for W is a separator $S \subseteq V_{\mathcal{C}}$ that separates the left from the right end of the grid $\mathcal{C}$ and satisfies additional conditions on the vertices that S is allowed to contain. For example, if W contains only one class of row i, then S may only contain a single vertex from that class. A set W is *t-critical*, if there is a t-virtual cordon for W and there is no periodic path satisfying the conditions of Lemma 21 that avoids all vertices of the classes in W (and actually even more). Intuitively, for t-critical sets the robber cannot move between both ends. For non-t-critical sets of size at most $k + t - 1$ (so in situations where at least one cop is picked up), this is always possible using periodic paths.

Lemma 22 ([19])**.** *Let $t \in [k-1]$ and let W be a set of at most $k + t - 1$ many $\equiv^t$-equivalence classes. If W is not t-critical, then there is a t-end-to-end-twisting avoiding all classes in W.*

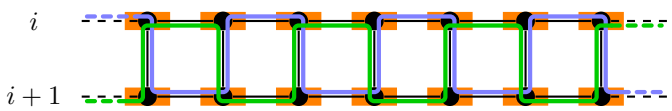


Figure 2: The two 2-periodic paths (in blue and in green) constructed in the proof of Lemma 23. Edges in and between rows i and $i+1$ in the cylindrical grid are drawn in black. Both paths never use the same incident edges of any vertex. Avoided horizontal roadblocks are drawn in orange.

The minimal distance of the robber to an inclusion-wise minimal t-virtual cordon for W measures the distance between the robber and the cops. When an announced cop will make the position t-critical, the robber moves to the end to which this distance is larger. This distance decreases by at most a constant in each round [19]. So, the robber still has a strategy for a number of rounds linear in the grid length. Since the compressed CFI graphs are much smaller, we get a much better bound for the k-pebble game on the compressed CFI graphs.

6.2 Cops do not Benefit From Roadblocks

We now show that for the $\mathcal{C}$-compressions of the previous section, the Cops Player does not benefit from roadblocks. This means that, although blocking moves possibly allow the Cops Player to make multiple moves per round, the number of rounds the robber can avoid getting caught does not change asymptotically compared to the game without roadblocks. Note that converting a roadblock to a cop only makes it harder for the robber to move. To see this, observe that a roadblock prevents the robber from passing through a vertex (or class) using a specified set of incident edges, while a cops prevents the robber from passing through the vertex (or class) at all.

Lemma 23. *Let $t \in [k-1]$ and $c \leq \frac{2}{5}k - 1$ be integers. Consider the compressed and blocking $(k+c)$-Cops and Robber game on $\mathcal{C}$ and $\equiv^t$ and assume that cops are placed in at most c rows. Then there is a t-end-to-end twisting that avoids all cop-occupied $\equiv^t$-equivalence classes and avoids all roadblocks.*

Proof Sketch. We call a roadblock *horizontal* if it blocks the use of exactly the two horizontal incident edges of a vertex or $\equiv^t$-class. If there are no horizontal roadblocks in a *cop-free* row, then the straight path through that row is a t-end-to-end twisting and we are done. Assume for a contradiction, that no t-end-to-end twisting exists. A cop-free row is *lonely*, if it is sandwiched by cop-occupied rows. We show that there have to be additional (non-horizontal) roadblocks in each non-lonely row. To do this we construct, for non-lonely rows i and $i+1$, two 2-periodic paths that only use vertices from rows i and $i+1$, avoid all horizontal roadblocks, and do not share the same incident edges of any vertex in rows i and $i+1$ (see Figure 2). By Lemma 21, if there are no non-horizontal roadblocks in rows i and $i+1$, then these path would induce t-end-to-end twistings. Because the two paths do not use common incident edges, an additional roadblock is required for each one to block the path. This allows us to lower bound the number of roadblocks in terms of c and to contradict the assumption that $c \leq \frac{2}{5}k - 1$. Hence, the desired t-end-to-end twisting exists. See Appendix D for the full proof. □

Using the previous lemma, we are ready to prove the main technical result of this section.

Lemma 24. *Let $1 \leq t \leq \frac{2}{5}k - 1$ be an integer. Then the robber, initially placed on the left or right end of $\mathcal{C}$, has a strategy for the first $\Omega(J)$ rounds in the compressed and blocking $(k+t)$-Cops and Robber game on $\mathcal{C}$ and $\equiv^t$.*

Proof Sketch. We will convert all roadblocks into cops and make the game harder for the robber. In this way, we use the notions of t-critical sets and t-virtual cordons for these positions. The robber always stays at one end of the grid: If the current position is not critical, the robber stays at the current end, until an announced cop or roadblock (seen as a cop) makes the position critical. Then using the t-end-to-end twisting from Lemma 22 the robber moves to the end of the grid with larger distance to every minimal t-virtual cordon. This distance is in $\Omega(J)$ [19]. If the current position is critical, we show that blocking moves only allow the Cops Player to decrease this distance by $O(k)$, via a case distinction on the number of cop-occupied rows. While this is at most $\frac{2}{5}k - 1$, the robber can always use the t-end-to-end twisting given by Lemma 23 to switch ends. Otherwise, the number of cop-occupied rows is at least $\frac{2}{5}k$. In this case, all intermediate positions between the blocking moves share at least one row in which only one and the same cop is placed, so by inductively applying [19, Proposition 4.10], we show that all minimal t-virtual cordon before and after the blocking moves are contained within $O(k)$ subsequent columns.

So, starting from a distance of $\Omega(J)$, the robber has a strategy such that this distance decreases by at most $O(k)$ in each round. Hence, the robber has a strategy for the first $\Omega(J/O(k)) = \Omega(J)$ rounds (since k is seen as a constant). The full proof is in Appendix D. □

Finally, for sufficiently large n and choosing $w = \lceil \sqrt[t+1]{n} \rceil$, Lemmas 24, 14, 19, and 18 together imply the desired round lower bound for the $(k+t)$-pebble game with blocking. See Appendix D for details.

Theorem 25. *For all integers $k \geq 3$, $1 \leq t \leq \frac{2}{5}k - 1$, and $n \in \mathbb{N}$, there are two colored graphs $\mathcal{G}$ and $\mathcal{H}$ of order $\Theta(n)$ and color class size 8 such that*

1. *$\mathcal{G} \not\simeq^{k+1} \mathcal{H}$, that is, Spoiler wins the $(k+1)$-pebble game on $\mathcal{G}, \mathcal{H}$, and*
2. *$\mathcal{G} \simeq_{\mathcal{B}}^{k+t,\Omega(n^{k/(t+1)})} \mathcal{H}$, that is, Duplicator has a strategy for the first $\Omega(n^{k/(t+1)})$ rounds in the $(k+t)$-pebble game with blocking on $\mathcal{G}, \mathcal{H}$.*

7 Supercritical Width versus Tree-Like Size Trade-Offs

We finally derive our main results; starting with narrow resolution.

Theorem 26. *For all integers $k \geq 3$, $1 \leq t \leq \frac{2}{5}k - 1$, and $n \in \mathbb{N}$, there are two colored graphs $\mathcal{G}$ and $\mathcal{H}$ of order $\Theta(n)$ and color class size 16 such that*

1. *there is a k-narrow resolution refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$, and*
2. *every $(k+t-1)$-narrow tree-like resolution refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ has size $2^{\Omega(n^{k/(t+1)})}$.*

Proof. Let $k \geq 3$ and $1 \leq t \leq \frac{2}{5}k - 1$. By Theorem 25, for all $n \in \mathbb{N}$, there are graphs $\mathcal{G}$ and $\mathcal{H}$ of color class size 8 and order $\Theta(n)$ such that $\mathcal{G} \not\simeq^{k+1} \mathcal{H}$ and $\mathcal{G} \simeq_{\mathcal{B}}^{k+t,\Omega(n^{k/(t+2)})} \mathcal{H}$. It is easy to see that $\mathcal{X}(\mathcal{G})$ and $\mathcal{X}(\mathcal{H})$ have color class size 16. By Theorem 5 and Lemma 6, there is a k-narrow resolution refutation for $\mathsf{ISO}(\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H}))$. Moreover, from Lemma 9 it follows that Delayer has a strategy to score $\Omega(n^{k/(t+1)})$ points in the $(k+t)$-narrow Prover-Delayer game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$. Therefore, by Lemma 4, the result follows. □

Theorem 1 is the case $t = 1$ of Theorem 26. We now lift Theorem 26 to usual resolution (without the narrow resolution rule). First, if $\mathcal{G}$ and $\mathcal{H}$ have color class size c, then we can convert every k-narrow refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ into a (usual) refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$ of width $k + c$. Second, a width-k refutation is in particular a width-k narrow refutation. **Theorem 2** follows immediately. Note that while Assertion 2 of Theorem 2 provides a lower bound for all $t \leq \frac{2}{5}k - 1$, Assertion 1 only guarantees a refutation of width $k + 16$. Therefore, the existing refutation must be large only for $k \geq 45$ and $17 \leq t \leq \frac{2}{5}k - 1$.

Conclusion and Open Questions. We established a new super-critical (narrow) width vs. tree-like size trade-off on graph isomorphism formulas for resolution. The lower bound of $2^{\Omega(n^{k/2})}$ obtained for $t = 1$ in Theorems 2 and 26 is close to the upper bound of 2^{n^k} for the tree-like size of resolution of (narrow) width k. We exploited a compressed variant of the CFI graphs and round number lower bounds in the k-pebble game on them. However, we had to move from the k-pebble game to the k-pebble game with blocking, and reprove the round number lower bounds in this setting. This raises the question of whether there is a generic translation from round number lower bounds in the k-pebble game to tree-like size resolution lower bounds. Another question is whether the decrease in the robustness in the trade-off from $2k$ in [19] to $\frac{7}{5}k$ in Theorem 26 is necessary. More broadly, we ask for a more robust compression or trade-off that can be applied to a much wider range than $2k$.

A Proof of Lemma 4

Lemma 27 (Lemma 4 restated)**.** *Let $k \geq 1$, let $\mathcal{G}, \mathcal{H}$ be colored graphs, and let π be a k-narrow refutation of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$. Then Prover has a $(\lceil \log(|\pi|) \rceil)$-point strategy in the $(k+1)$-narrow Prover-Delayer game on $\mathcal{G}, \mathcal{H}$.*

Proof. Let k, $\mathcal{G}, \mathcal{H}$ and π be as in the statement of the lemma. The proof proceeds by showing that Prover can use π to construct a winning strategy in the $(k+1)$-narrow Prover-Delayer game on $\mathcal{G}, \mathcal{H}$. To do this, Prover traverses π starting from the empty clause, which is its root. If in round r Prover is at some clause of π, then in the next round Prover moves either to a child of this clause or to a leaf while maintaining a certain invariant. In order to state the invariant we require the following notion. A partial assignment σ *contradicts* a clause $C \in \pi$ if

1. for every negated variable $\neg x_{u,v} \in C$, we have $\sigma(x_{u,v}) = 1$; and
2. for every positive variable $x_{u,v} \in C$ either
   (a) $\sigma(x_{u,v}) = 0$ or
   (b) there is some $x_{u',v'} \in \mathsf{dom}(\sigma)$ with $\sigma(x_{u',v'}) = 1$ such that $(\neg x_{u,v} \vee \neg x_{u',v'}) \in \mathsf{ISO}(\mathcal{G}, \mathcal{H})$.

We show by induction of the number of rounds, that Prover has a strategy such that after round t the following invariant holds:

(I) If Delayer has p points, then the partial assignment σ_t contradicts a clause $C \in \pi$ and the subtree of π rooted at C has size at most $\lfloor |\pi|/2^p \rfloor$. Moreover, C is not a color clause of $\mathsf{ISO}(\mathcal{G}, \mathcal{H})$.

First, suppose that Prover can maintain (I) and that in round t Delayer scores a point to take the total score to $\lceil \log(|\pi|) \rceil$. We show that now the game ends. Observe that by (I), the assignment σ_t contradicts a clause C such that the subtree of π rooted at C has size one. Therefore, $C \in \mathsf{ISO}(\mathcal{G}, \mathcal{H})$. Moreover, again by (I), C is either a bijection or an edge clause. Since such clauses contain only negated variables, σ_t violates C. Therefore, the game ends in round t. It follows that if Prover maintains (I), then Delayer can score at most $\lceil \log(|\pi|) \rceil$ many points. It remains to show that Prover indeed can maintain the invariant.

Before the first round, Delayer has zero points and σ_0 trivially violates the empty clause, which is the root of π. This acts as our base case. So suppose that for some $t \geq 1$ the invariant holds at the end of round $t-1$. If σ_{t-1} violates a leaf, then—as previously argued—the game ends. Otherwise, there is some $C \in \pi$ such that:

1. σ_{t-1} contradicts C but does not contradict any C' lying in the subtree of π rooted at C, and
2. the subtree of π rooted at C has size at most $\lfloor |\pi|/2^p \rfloor$.

Prover begins by choosing $\sigma \subseteq \sigma_{t-1}$ to be the assignment including only the variables needed to contradict C. This implies $|\sigma| = w(C) \leq k$, since $C \notin \mathsf{ISO}(\mathcal{G}, \mathcal{H})$. We know that C is derived by either the resolution or the narrow resolution rule; we deal with each of these possibilities in turn. To simplify the case analysis, we assume, without loss of generality, that no color clause is ever an assumption in an application of the resolution rule: such steps can be replaced by narrow resolution steps. We further assume, again without loss of generality, that every narrow resolution step *not* involving a color clause has exactly three assumptions: if not we can replace such steps with resolution steps.

Case 1: Resolution. Suppose C is derived from $C' = A \vee x_{u,v}$ and $C'' = B \vee \neg x_{u,v}$, where $x_{u,v}$ is the resolved variable. Suppose further that $x_{u,v} \notin \mathsf{dom}(\sigma)$. Then Prover makes a resolution move and chooses $x_{u,v}$. Now we observe that since σ contradicts C:

- $\sigma[x_{u,v} \mapsto 0]$ contradicts C' and
- $\sigma[x_{u,v} \mapsto 1]$ contradicts C''.

Recall that by assumption neither C' nor C'' is a color clause. If Delayer makes a committal move, then σ_t either contradicts C' or C''. Moreover, the subtrees rooted at both these clauses are smaller than the one rooted at C and so (I) is maintained. Otherwise, Delayer makes a point move. Then Prover can choose the value of $\sigma_t(x_{i,j})$, Therefore, Prover can in effect choose which of C' and C'' is contradicted by σ_t. Prover chooses the one with the smaller subtree in π. This is at most half the size of the subtree rooted at C and so (I) is again maintained.

So suppose otherwise that $x_{u,v} \in \mathsf{dom}(\sigma)$. If $\sigma(x_{u,v}) = 0$, then σ contradicts C'. If $\sigma(x_{u,v}) = 1$, then σ contradicts C''. But by the choice of C, neither of these situations occur. Hence, $x_{u,v} \notin \mathsf{dom}(\sigma)$. Therefore, Prover can maintain (I) if C is derived by resolution.

Case 2: Narrow Resolution. If C is derived by narrow resolution, then some clause of $\mathsf{ISO}(\mathcal{G},\mathcal{H})$ is an assumption in the derivation of C. We need to distinguish two subcases.

Case 2a: A color clause is an assumption. Let D be this assumption. Prover now queries D. Let $x_{u,v} \in D \setminus \sigma^{-1}(0)$. If $x_{u,v}$ is resolved at this step, then there is some assumption C' of the form $(A \vee \neg x_{u,v})$. Since σ contradicts C, the assignment $\sigma[x_{u,v} \mapsto 1]$ contradicts C'. If $x_{u,v}$ is not resolved at this step, then it occurs positively in C. Since σ contradicts C and as $x_{i,j} \notin \sigma^{-1}(0)$ by assumption, there is some variable $x_{u',v'}$ such that $\sigma(x_{u',v'}) = 1$ and $C'' := (\neg x_{u,v} \vee \neg x_{u',v'}) \in \mathsf{ISO}(\mathcal{G},\mathcal{H})$. Therefore, $\sigma[x_{u,v} \mapsto 1]$ violates C''.

From this it is easy to see that Prover can maintain the invariant similarly to the resolution case. In detail, suppose that Delayer makes a committal moves and chooses $x \in D \setminus \sigma^{-1}(0)$. Then, $\sigma_t = \sigma[x \mapsto 1]$ either violates a bijection or an edge clause—so the game ends—or contradicts an assumption used to derive C. Either way, (I) is maintained. Suppose instead that Prover makes a point move and chooses $\{x, y\}$. If, for some $z \in \{x, y\}$, the assignment $\sigma[z \mapsto 1]$ violates a bijection or an edge clause of $\mathsf{ISO}(\mathcal{G},\mathcal{H})$, then Prover chooses this z and the game ends. Otherwise, by the analysis above, $\sigma[x \mapsto 1]$ and $\sigma[y \mapsto 1]$ contradict distinct children of C. Via their choice of z, Prover can decide which of these two children are contradicted and so—similarly to in the resolution case—can maintain (I).

Case 2b: No color clause is an assumption. In this case, there is some assumption $D = (\neg x_{u,v} \vee \neg x_{u',v'})$ which is either a bijection or an edge clause. Moreover, there are three assumptions in total and so the other two must be of the form $(A \vee x_{u,v})$ and $(B \vee x_{u',v'})$. If $x_{u,v}$ occurs positively in C, then $\sigma(x_{u,v}) = 0$ and so σ contradicts $(A \vee x_{u,v})$. But this contradicts our choice of C, since $(A \vee x_{u,v})$ is in the subtree of π rooted at C. Therefore, $x_{u,v} \notin \sigma^{-1}(0)$. Moreover, if $\sigma(x_{u,v}) = 1$, then σ contradicts $(B \vee x_{u',v'})$ since $D = (\neg x_{u,v} \vee \neg x_{u',v'})$ is either a bijection or an edge clause. Again, by the choice of C, this case does not occur. Hence, $x_{u,v} \notin \mathsf{dom}(\sigma)$. Therefore, Prover makes a resolution move and selects $x_{u,v}$. Observe that:

- $\sigma[x_{u,v} \mapsto 0]$ contradicts $(A \vee x_{u,v})$ and
- $\sigma[x_{u,v} \mapsto 1]$ contradicts $(B \vee x_{u',v'})$.

Therefore, by essentially the same argument as when C is derived by resolution, Prover can maintain (I). □

B Proofs from Section 4

Lemma 28 (Lemma 6 restated)**.** *Let $k \geq 3$ and $\mathcal{G}$ and $\mathcal{H}$ be colored graphs that do not have connected twins. If $\mathcal{G} \not\simeq^{k,r} \mathcal{H}$, then $\mathcal{X}(\mathcal{G}) \not\simeq^{k,r+1} \mathcal{X}(\mathcal{H})$.*

Proof. By Theorem 5, it suffices to show that if Spoiler has a winning strategy in the r-round k-pebble game on $\mathcal{G},\mathcal{H}$, then Spoiler as a winning strategy in the $(r+1)$-round k-pebble game on $\mathcal{X}(\mathcal{G}),\mathcal{X}(\mathcal{H})$. The following claim is crucial.

Claim 29. *Let some $p, q \in [k]$ and (α, β) be a position in the k-pebble game on $\mathcal{X}(\mathcal{G}),\mathcal{X}(\mathcal{H})$ such that $\{\alpha(p), \alpha(q)\} = \mathcal{X}_{\mathcal{G}}(v)$, for some $v \in V_{\mathcal{G}}$, and $\{\beta(p), \beta(q)\} \neq \mathcal{X}_{\mathcal{H}}(u)$, for every $u \in V_{\mathcal{H}}$. Then Spoiler has a winning strategy in one round from position (α, β).*

Proof of claim. Let $x := \beta(p)$ and $y := \beta(q)$. First note that if $x = y$, then the game ends immediately. Otherwise, $u := \mathcal{X}_{\mathcal{H}}^{-1}(x) \neq \mathcal{X}_{\mathcal{H}}^{-1}(y) =: v$. Suppose $\{u, v\} \notin E_{\mathcal{H}}$. Then $\{x, y\} \notin E_{\mathcal{X}(\mathcal{H})}$ so—since there is an edge between $\alpha(p)$ and $\alpha(q)$—the game again ends immediately. If $\{u, v\} \in E_{\mathcal{H}}$, then by assumption u and v are not twins. Let $w \in V_{\mathcal{H}}$ be a witness for this, i.e., w is either adjacent to u and not v or vice-versa. In the next round, Spoiler picks the o-th pebble pair for some $o \notin \{q, p\}$. This is possible since $k \geq 3$. Spoiler then places the o-th pebble for $\mathcal{X}(\mathcal{H})$ on $w_0 \in V_{\mathcal{X}(\mathcal{H})}$. Since $\alpha(p)$ and $\alpha(q)$ are connected twins, every response of Duplicator is immediately losing. □

We associate a position $P = (\alpha, \beta)$ in the k-pebble game on $\mathcal{X}(\mathcal{G}),\mathcal{X}(\mathcal{H})$, with a position $\mathcal{X}^{-1}(P) := (\gamma, \delta)$ in the k-pebble game on $\mathcal{G},\mathcal{H}$ via $\gamma(i) := \mathcal{X}^{-1}(\alpha(i))$ and $\delta(i) := \mathcal{X}^{-1}(\beta(i))$ for all $i \in [k]$. We show by induction on the number of rounds t that if Spoiler can win in t rounds from $\mathcal{X}^{-1}(P)$, then Spoiler can win from position P in at most $t + 1$ rounds.

For the base case $t = 0$, Spoiler immediately wins in position $\mathcal{X}^{-1}(P)$ and we need to prove that Spoiler wins in ≤ 1 round from P. First suppose there are distinct $p, q \in [k]$ such that $\gamma(p) = \gamma(q)$ or $\delta(p) = \delta(q)$ but $\delta(p) \neq \delta(q)$ or $\gamma(p) \neq \gamma(q)$, respectively. Then Spoiler can win from P in at most one round by Claim 29. Otherwise, there are distinct $p, q \in [k]$ such that $\{\gamma(p), \gamma(q)\} \in E_{\mathcal{G}}$ if and only if $\{\delta(p), \delta(q)\} \notin E_{\mathcal{H}}$. But then $\{\alpha(p), \alpha(q)\} \in E_{\mathcal{X}(\mathcal{G})}$ if and only if $\{\beta(p), \beta(q)\} \notin E_{\mathcal{X}(\mathcal{H})}$. So P is a winning position for Spoiler.

For the inductive step, suppose that Spoiler has a strategy to win in $t+1$ rounds from $\mathcal{X}^{-1}(P)$. Suppose that the next move of Spoiler according to this strategy is to pick up the p-th pebble pair and to place the p-th pebble for $\mathcal{G}$ on v. In the game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$, we stipulate that Spoiler also picks up the p-th pebble pair and place the p-th pebble for $\mathcal{G}$ on v_0. Duplicator responds by placing the p-th pebble for $\mathcal{X}(\mathcal{H})$ on some $u_i \in V_{\mathcal{X}(\mathcal{H})}$ for $i \in \{0,1\}$. Let P' be the resulting position in the game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$. In the game on $\mathcal{G}, \mathcal{H}$, we let Duplicator respond by placing the p-th pebble on u. Let Q be the resulting position in the game on $\mathcal{G}, \mathcal{H}$. Clearly, $Q = \mathcal{X}^{-1}(P')$. Since we followed the winning strategy for Spoiler in the game on $\mathcal{G}, \mathcal{H}$, Spoiler has a winning strategy in t rounds from Q. By the inductive hypothesis, Spoiler has also a winning strategy in $t+1$ rounds from P'. Hence, Spoiler has a winning strategy from P in the game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$ in $t+2$ rounds. The case where Spoiler places a pebble on $\mathcal{H}$ in position $\mathcal{X}^{-1}(P)$ is analogous. □

Lemma 30 (Lemma 9 restated)**.** *Let $\mathcal{G}$ and $\mathcal{H}$ be colored graphs and $k \geq 2$ an integer. If Duplicator has a winning strategy for the r-round k-pebble game with blocking on $\mathcal{G}, \mathcal{H}$, then Delayer has an r-point strategy in the k-narrow Prover-Delayer game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$.*

Proof. The intuition behind Delayer's strategy for the k-narrow Prover-Delayer game played on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$ is to simulate in the background positions of the k-pebble game with blocking on $\mathcal{G}$ and $\mathcal{H}$. Whenever Duplicator marks a pebble pair placed on vertices u and v as $\mathsf{regular}$ in the game (and there is not already a pebble pair on u and v), Delayer should score a point in the Prover-Delayer game since it 'does not matter' whether we map u_0 to v_0 or to v_1. But this is only the case if no existing variable assignments fix the image of u_0. For instance if x_{u_1,v_1} is already assigned zero, then this forces u_0 to be mapped to v_1. However, whenever this is the case, we can show that Delayer already scored a point for this earlier assignment.

We formalize this idea. Let σ be a position in the k-narrow Prover-Delayer game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$. A *point variable in position σ* is a variable $x_{u,v} \in \sigma^{-1}(0)$ such that for all variables $x_{u',v'} \in \mathsf{dom}(\sigma)$ with $\mathcal{X}^{-1}(u') = \mathcal{X}^{-1}(u)$ and $\mathcal{X}^{-1}(v') = \mathcal{X}^{-1}(v)$, we have $u = u'$ and $v = v'$. Intuitively, point variables are those critical zero assignments—as in the previous example—that force Delayer to set a variable to one and for which a point was already scored.

A position (α, β, c) in the k-pebble game with blocking on $\mathcal{G}$ and $\mathcal{H}$ is a *t-round witness* for a pair (σ, V) of a position σ in the k-narrow Prover-Delayer game and a set of point variables $V \subseteq \sigma^{-1}(0)$ in position σ if

- there is a variable $x_{u,v} \in \sigma^{-1}(1)$ if and only if there is a pebble pair $p \in [k]$ with $c(p) = \mathsf{regular}$, $\alpha(p) = \mathcal{X}^{-1}(u)$, and $\beta(p) = \mathcal{X}^{-1}(v)$,
- there is a variable $x_{u,v} \in \sigma^{-1}(0) \setminus V$ if and only if there is a pebble pair $p \in [k]$ such that $\alpha(p) = \mathcal{X}^{-1}(u)$, $\beta(p) = \mathcal{X}^{-1}(v)$, and $c(p) = \mathsf{blocking}$, and
- Duplicator has a winning strategy for the t-round game starting in position (α, β, c).

Claim 31. *For all $t \in \mathbb{N}$, $\ell \leq k$, all positions σ in the k-narrow the Prover-Delayer game on $\mathcal{X}(\mathcal{G}), \mathcal{X}(\mathcal{H})$, and all sets $V \subseteq \sigma^{-1}(0)$ of point variables in σ of size ℓ, if (σ, V) has a $(t+\ell)$-round witness, then Delayer has a t-point strategy from σ.*

Proof of claim. We proceed by induction on t. For the base $t = 0$, note that trivially Delayer has a 0-point strategy from any position. So assume that the inductive hypothesis holds for some t. Let $\ell \in [k]$. We now do a further induction on ℓ. Since the base and the inductive case are quite similar, we will not split them but argue that the steps where we use the inductive hypothesis for ℓ cannot occur if $\ell = 0$. Let σ be a position in the k-narrow Prover-Delayer game and $V \subseteq \sigma^{-1}(0)$ be a set of point variables in σ of size ℓ. Suppose that (σ, V) has a $(t+1+\ell)$-round witness P. Prover chooses a subset $\sigma' \subseteq \sigma$ of size at most $k-1$. We pick up the corresponding pebble pairs in P, yielding the position P'. Clearly P' is a $(t+1+\ell)$-round witness for $(\sigma', V \cap \mathsf{dom}(\sigma'))$. First suppose that $V \not\subseteq \mathsf{dom}(\sigma')$. In this case ℓ is nonzero and by the induction hypothesis for ℓ and since $|V \cap \mathsf{dom}(\sigma')| < \ell = |V|$, Delayer has a $(t+1)$-point strategy from σ' and therefore from σ. Now we can assume that $V \subseteq \mathsf{dom}(\sigma')$ so that P' is a $(t+1+\ell)$-round witness for (σ', V). We make a case distinction on the action of Prover.

Case 1: Resolution move. Prover chooses a variable $x_{u,v} \notin \mathsf{dom}(\sigma)$. To determine Delayer's response, we suppose that from position P' in the k-pebble game with blocking Spoiler makes a blocking move and places a pebble pair on $\mathcal{X}^{-1}(u)$ in $\mathcal{G}$ and $\mathcal{X}^{-1}(v)$ in $\mathcal{H}$. Since P' is a $(t+1+\ell)$-round witness for (σ', V), it follows that Duplicator has a winning strategy in the $(t+1+\ell)$-round k-pebble game with blocking starting from position P'. We assume that Duplicator responds according to such a strategy. First suppose that Duplicator responds by marking this pebble pair with $\mathsf{blocking}$ yielding position P''. We let Delayer make a committal response and set $x_{u,v}$ to be zero yielding position σ''. It is easy to see, that P'' is a $(t+1+\ell)$-round witness for (σ'', V).

By induction on the number of blocking rounds in which Duplicator responds by marking the pebble pair placed as blocking, we can repeat this argument until either Duplicator answers by marking the pebble pair placed as regular, or Spoiler plays a regular move. Assume that, when this situation finally occurs, the k-pebble game with blocking is in position Q, and the Prover-Delayer game is in position τ. By the inductive argument, Q is a $(t+1+\ell)$-round witness for (τ, V). By assumption, from Q a new position Q' is reached by placing a new pebble pair marked regular on $\mathcal{X}^{-1}(u)$ in $\mathcal{G}$ and on $\mathcal{X}^{-1}(v)$ in $\mathcal{H}$; note that it does not matter whether this happens during a blocking move or a regular move. Moreover, Duplicator has a winning strategy in the $(t+\ell)$-round k pebble game with blocking starting from position Q'. There are two possibilities to consider.

- If there is a variable $x_{u',v'} \in V$ such that $\mathcal{X}^{-1}(u') = \mathcal{X}^{-1}(u)$ and $\mathcal{X}^{-1}(v') = \mathcal{X}^{-1}(v)$, then Delayer makes a committal response and sets $x_{u,v}$ to be one. This yields position τ'. Set $V' := V \setminus \{x_{u,v}\}$. Hence, $|V'| = \ell - 1$. Now, the position Q' is a $(t+\ell)$-round witness for (τ', V'). Note that in this case $\ell \neq 0$ so we may apply the induction hypothesis for ℓ. Since $t+\ell = (t+1)+|V'|$, it follows that Delayer has a $(t+1)$-point strategy from τ' and hence from σ.

- Otherwise, there is no such variable in V. In this case, Delayer makes a point response and Prover assigns a value to $x_{u,v}$ resulting in position τ'. If $\tau'(x_{u,v}) = 1$, then we set $V' := V$. The position Q' is a $(t+\ell)$-round witness for (τ', V'). By the inductive hypothesis for t, Delayer has a t-point strategy from τ' and so has a $(t+1)$-strategy from σ.

  If instead $\tau'(x_{u,v}) = 0$, then $x_{u,v}$ is a point variable. Therefore we set $V' := V \cup \{x_{u,v}\}$. Hence, $|V'| = |V| + 1$ and Q is a $(t+1+\ell) = (t+|V'|)$-round witness for (τ', V')—note that here we consider Q instead of Q' because we add $x_{u,v}$ to the point variables and we hence want to 'forget' the pebble pair placed in this round. Thus, by the inductive hypothesis for t, Delayer has a t-point strategy from position τ'. Since Delayer scored a point in this round, Delayer has a $(t+1)$-point strategy from position σ.

Case 2: Narrow move. Prover chooses a color clause. Assume there is some $u \in V_{\mathcal{X}(\mathcal{G})}$ such that this clause is $\bigvee_{v \in W_u} x_{u,v}$ where $W_u = \chi_{\mathcal{H}}^{-1}(\chi_{\mathcal{G}}(u))$. The case that u is picked from $V_{\mathcal{X}(\mathcal{H})}$ is symmetric. To determine Delayer's response, we suppose that Spoiler makes a regular move from position P' and places an unused pebble on $\mathcal{X}^{-1}(u)$ on $\mathcal{G}$. Again, since P' is a $(t+1+\ell)$-round witness for (σ', V), Duplicator has a winning strategy in the $(t+1+\ell)$-round k-pebble game with blocking starting from position P'. Suppose that Duplicator answer with a vertex $v \in V_{\mathcal{H}}$ according to such a strategy and denote the resulting position in the k-pebble game with blocking by P''. We distinguish three cases:

- Assume that $x_{u,v_0} \in (\sigma')^{-1}(0)$ or $x_{u,v_1} \in (\sigma')^{-1}(0)$. Since both cases are analogous, we assume $x_{u,v_0} \in (\sigma')^{-1}(0)$. We claim that in this case $x_{u,v_0} \in V$. To see this, suppose for a contradiction $x_{u,v_0} \notin V$. Then in position P', there is a pair of blocking pebbles lying on $\mathcal{X}^{-1}(u)$ in $\mathcal{G}$ and on v in $\mathcal{H}$. This implies that Spoiler wins in position P'', which is a contradiction to the fact that P is a $(t+1+\ell)$-witness for (σ, V) and that we followed the strategy of Duplicator to obtain position P''. Hence, $x_{u,v_0} \in V$ and so x_{u,v_0} is in particular a point variable in position σ. It follows that $x_{u,v_1} \notin \mathsf{dom}(\sigma)$. In this case, Delayer makes a committal response and chooses x_{u,v_1}, yielding position σ''. We set $V'' := V \setminus \{x_{u,v_0}\}$. Then P'' is a $(t+\ell) = ((t+1)+|V''|)$-round witness for (σ'', V''). In this case $\ell \neq 0$ so we may apply the inductive hypothesis for ℓ. Therefore, Delayer has a $(t+1)$-point strategy from position σ'' and thus from position σ.

- Assume that $x_{u,v_0} \in (\sigma')^{-1}(1)$ or $x_{u,v_1} \in (\sigma')^{-1}(1)$. We consider the case $x_{u,v_0} \in \sigma^{-1}(1)$, the other one is analogous. Delayer makes a committal response and chooses x_{u,v_0}. We see that in this case the Prover-Delayer game did not make any progress at all since $\sigma(x_{u,v_0}) = 1$. Hence, we may assume that this case does not occur.

- It remains to consider the case $\{x_{u,v_0}, x_{u,v_1}\} \cap \mathsf{dom}(\sigma) = \emptyset$. We set $V'' := V$. Delayer makes a point response and chooses $\{x_{u,v_0}, x_{u,v_1}\}$. Spoiler sets one of these two variables to one yielding position σ''. The position P'' is a $(t+\ell) = (t+|V''|)$-round witness for (σ'', V''). By the inductive hypothesis for t, Delayer has a t-point strategy from position σ''. Since a point was scored in this round, Delayer has a $(t+1)$-point strategy from σ. □

Finally we show that the claim implies the lemma. The k-narrow Prover-Delayer game starts from the empty position. Moreover, $(\emptyset, \emptyset)$ clearly has an r-round witness since Duplicator has a winning strategy in the r-round k-pebble game with blocking on $\mathcal{G}$ and $\mathcal{H}$. Hence, the claim implies that Delayer has an r-point strategy from the initial position. □

C Proofs from Section 5

Lemma 32 (Lemma 14 restated)**.** *Let $k \geq 3$, $r \in \mathbb{N}$, let $\equiv$ be a $\mathcal{G}$-compression, and let $f, g : E_{\mathcal{G}} \to \mathbb{F}_2$ be $\equiv$-compressible.*

1. $\mathsf{CFI}(\mathcal{G}, f) \not\simeq_{\mathcal{B}}^{k,r} \mathsf{CFI}(\mathcal{G}, g)$ *implies* $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \not\simeq_{\mathcal{B}}^{k,r} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$.
2. $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \not\simeq_{\mathcal{B}}^{k,r} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$ *implies* $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv} \not\simeq_{\mathcal{B}}^{k,r} \mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$.
3. $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv} \not\simeq_{\mathcal{B}}^{k,r} \mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$ *implies* $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \not\simeq_{\mathcal{B}}^{k,r+2} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$.

Proof. The following proof is a rather straight-forward adaption of the proof of the corresponding statement for the bijective k-pebble game from [25].

1. Trivial since $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ just extend $\mathsf{CFI}(\mathcal{G}, f)$ and $\mathsf{CFI}(\mathcal{G}, g)$, respectively, by an additional relation.

2. Assume that $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \not\simeq_{\mathcal{B}}^{k,r} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$, that is, Spoiler has a winning strategy in the r-round k-pebble game with blocking played on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$. We show that Spoiler also has a winning strategy in the r-round game on $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv}$ and $\mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$.

   First note that we can always assume the Duplicator plays color preserving, which means that Duplicator always answers for regular pebbles with a vertex of the same color. Otherwise, Duplicator would lose immediately.

   Consider a position (α, β, c) of the game on $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv}$ and $\mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$. We say that a position (α', β', c') of the game on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ is an *s-round witness* for (α, β, c) if the following conditions are satisfied:

   (a) $\mathsf{dom}(\alpha) = \mathsf{dom}(\beta) = \mathsf{dom}(c) = \mathsf{dom}(\alpha') = \mathsf{dom}(\beta') = \mathsf{dom}(c')$;

   (b) $c(i) = c'(i)$ for all $i \in \mathsf{dom}(c)$;

   (c) $\alpha'(i)/_{\equiv} = \alpha(i)$ and $\beta'(i)/_{\equiv} = \beta(i)$ for all $i \in \mathsf{dom}(\alpha)$;

   (d) $\alpha'(i)$ has the same color in $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ as $\beta'(i)$ has in $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ for every $i \in \mathsf{dom}(\alpha)$ such that $c'(i) = \mathsf{regular}$;

   (e) (α', β', c') is a winning position for Spoiler in the s-round game on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$.

   We first show that if (α', β', c') is a 0-round witness for (α, β, c), then Spoiler wins the game on $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv}$ and $\mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$. Because (α', β', c') is a winning position for Spoiler in the 0-round game on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ by Condition (e), the map $\alpha'(i) \mapsto \beta'(i)$ for all i with $c'(i) = \mathsf{regular}$ is not a partial isomorphism or does not respect the blocking pebbles. We show that the mapping $\alpha'(i) \mapsto \beta'(i)$ for all i with $c(i) = \mathsf{regular}$ is not a partial isomorphism or does not respect the blocking pebbles.

   First note that $\alpha(i)$ has the same color as $\beta(i)$ and $\alpha'(i)$ has the same color as $\beta'(i)$ for every $i \in \mathsf{dom}(\alpha)$ with $c(i) = c'(i) = \mathsf{regular}$ because Duplicator plays color-preserving and because of Condition (d).

   Now assume that $\alpha'(i) \mapsto \beta'(i)$ does not respect the blocking pebbles, i.e. that there are $i, j \in [k]$ with $c'(i) = \mathsf{regular}$, $c'(j) = \mathsf{blocking}$, and $(\alpha'(i), \beta'(i)) = (\alpha'(j), \beta'(j))$. Then, by Condition (c), we have that $(\alpha(i), \beta(i)) = (\alpha'(i)/_{\equiv}, \beta'(i)/_{\equiv}) = (\alpha'(j)/_{\equiv}, \beta'(j)/_{\equiv}) = (\alpha(j), \beta(j))$. Because we have $c(i) = \mathsf{regular}$ and $c(j) = \mathsf{blocking}$ by Condition (b), the position (α, β, c) also does not respect the blocking pebbles and hence Spoiler wins immediately. So we may assume that the mapping respects the blocking pebbles.

   Next, suppose that $\alpha'(i) \mapsto \beta'(i)$ for all i with $c'(i) = \mathsf{regular}$ is not a partial isomorphism. If there exists $i, j \in [k]$, $c'(i) = c(j) = \mathsf{regular}$ such that $\alpha'(i) \equiv \alpha'(j)$ but $\beta'(i) \not\equiv \beta'(j)$, then by Condition (c), we have $\alpha(i) = \alpha(j)$ but $\beta(i) \neq \beta(j)$. Spoiler wins immediately. If instead $\alpha'(i) = \alpha'(j)$ but $\beta'(i) \neq \beta'(j)$, then—since Duplicator plays color-preserving—$\beta'(i)$ has the same color as $\beta'(j)$ (namely the one of $\alpha'(i)$) and $\beta'(i) \not\equiv \beta'(j)$ because distinct vertices of the same color are never $\equiv$-equivalent. Thus, $\alpha(i) = \alpha(j)$ but $\beta(i) \neq \beta(j)$ by Condition (c). Spoiler again wins immediately.

   Otherwise, the map does not respect the edge relation. So suppose that $\{\alpha'(i), \alpha'(j)\}$ is an edge in $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ but $\{\beta'(i), \beta'(j)\}$ is not an edge in $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$. Then $\{\alpha(i), \alpha(j)\}$ is an edge in $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv}$. We prove that $\{\beta(i), \beta(j)\}$ is not an edge in $\mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$ and hence that Spoiler wins the game on $\mathsf{CFI}(\mathcal{G}, f)/_{\equiv}$, $\mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$. Because $\{\alpha'(i), \alpha'(j)\}$ is an edge, the origins of $\alpha'(i)$ and $\alpha'(j)$ are adjacent base vertices. Hence, the origins of $\beta'(i)$ and $\beta'(j)$ are also adjacent base vertices, because they have the same colors as $\alpha'(i)$ and $\alpha'(j)$, respectively. Condition 2 of a graph compression implies that the set $\{u, v\}$ is not an edge in $\mathsf{CFI}(\mathcal{G}, g)/_{\equiv}$ (see [26, Lemma 11]).

The same arguments apply if $\{\beta'(i), \beta'(j)\}$ is an edge in $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ but $\{\alpha'(i), \alpha'(j)\}$ is not an edge in $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$; this completes the proof that if (α', β', c') is a 0-round witness for (α, β, c), then Spoiler wins the game on $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$.

It remains to prove, by induction on $s \leq r$, that Spoiler has a strategy for the r-round game on $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$ such that the position in round s has an $(r-s)$-round witness.

The base case $s = 0$ holds because the empty position in the game on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ is an r-round witness for the empty position in the game on $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$.

For the inductive step, let $s < r$ and suppose that the position in round s has an $(r-s)$-witness. We show by induction on m, that after m blocking moves (but no regular ones), the resulting position still has an $(r-s)$-witness.

The case $m = 0$ is trivial. So assume that the position (α, β, c) after m moves still has an $(r-s)$-witness (α', β', c'). Assume that, on the precompressed graphs $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$, Spoiler plays a blocking move. If, in the game on the precompressed CFI graphs, Spoiler removes a pair of pebbles, then Spoiler picks up the same pebble pair in the game on the compressed CFI graphs. We can remove the corresponding elements for the position and the witness relation remains intact. Now suppose Spoiler wants to place a pebble pair, say the ℓ-th. Let $u \in V_{(\mathsf{CFI}(\mathcal{G}, f), \equiv)}$ and $v \in V_{(\mathsf{CFI}(\mathcal{G}, g), \equiv)}$ be the vertices on which Spoiler places the ℓ-th pebble pair. In the game on the compressed CFI graphs $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$, we stipulate that Spoiler plays a blocking move and places the ℓ-th pebble pair on $u/_\equiv$ and $v/_\equiv$.

If Duplicator now marks this pair with $\mathsf{regular}$, then we are done since there is nothing to prove. So assume Duplicator marks the pair with $\mathsf{blocking}$. Then the resulting position is $(\alpha'[\ell \mapsto u], \beta'[\ell \mapsto v], c'[\ell \mapsto \mathsf{blocking}])$. This is an $(r-s)$-witness for the position $(\alpha[\ell \mapsto u/_\equiv], \beta[\ell \mapsto v/_\equiv], c[\ell \mapsto \mathsf{blocking}])$ after $m+1$ blocking moves since Spoiler has a strategy to win on the precompressed CFI graphs.

Finally, we show that after a regular move the resulting position has an $(r-s-1)$-witness. Picking up a pebble pair from the precompressed CFI graphs is transferred to the compressed CFI graphs as before. Now suppose Spoiler wants to place a pebble on the precompressed CFI graphs, say the ℓ-th. Assume that Spoiler places the pebble on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ (the case for $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ is analogous). Let $u \in V_{(\mathsf{CFI}(\mathcal{G}, f), \equiv)}$ be the vertex on which Spoiler places the pebble. Then in the game on the compressed CFI graphs $V_{\mathsf{CFI}(\mathcal{G}, f)/_\equiv}$ and $V_{\mathsf{CFI}(\mathcal{G}, g)/_\equiv}$, Spoiler places a pebble on $u/_\equiv$. Assume that Duplicator answers with $v/_\equiv$. Because we can assume that Duplicator plays color respecting, there is a unique vertex $w \in v/_\equiv$ that has the same color as u. We let Duplicator respond with w in the game on the the precompressed CFI graphs. Then the resulting position $(\alpha'[\ell \mapsto u], \beta'[\ell \mapsto w], c'[\ell \mapsto \mathsf{regular}])$, is an $(r-s-1)$-witness for the new position $(\alpha[\ell \mapsto u/_\equiv], \beta[\ell \mapsto v/_\equiv], c[\ell \mapsto \mathsf{regular}])$.

3. Assume that $\mathsf{CFI}(\mathcal{G}, f)/_\equiv \not\simeq^{k,r}_{\mathcal{B}} \mathsf{CFI}(\mathcal{G}, g)/_\equiv$. We turn a Spoiler winning strategy in the r-round k-pebble game with blocking played on $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$ into a Spoiler winning strategy in the $(r+2)$-round k-pebble game with blocking played on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$.

   A position (α', β', c') in the game played on $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$ is an *s-witness* for a position (α, β, c) in the game played on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ if

   (a) $\mathsf{dom}(\alpha) = \mathsf{dom}(\beta) = \mathsf{dom}(c) = \mathsf{dom}(\alpha') = \mathsf{dom}(\beta') = \mathsf{dom}(c')$;

   (b) $c(i) = c'(i)$ for all $i \in \mathsf{dom}(c)$;

   (c) $\alpha'(i) = \alpha(i)/_\equiv$ and $\beta'(i) = \beta(i)/_\equiv$ for all $i \in \mathsf{dom}(\alpha)$;

   (d) Spoiler has a winning strategy in the next s rounds in position (α, β', c').

   We first show that if (α, β, c) has a 0-witness (α', β', c'), then Spoiler wins the game on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ in at most 2 additional rounds. Since (α', β', c') is a 0-witness, Spoiler wins in this position. First, suppose the position does not respect the blocking pebbles. So there are $i, j \in [k]$ with $c'(i) = \mathsf{regular}$, $c'(j) = \mathsf{blocking}$, and $(\alpha'(i), \beta'(i)) = (\alpha'(j), \beta'(j))$. Then, by Condition (c), we have that $\alpha(i) \equiv \alpha(j)$ and $\beta(i) \equiv \beta(j)$. Spoiler picks up any pebble pair apart from the i-th and j-th pair, say the ℓ-th one (such a pair exists because $k \geq 3$). Spoiler plays a regular move and picks $\alpha(j)$. If Duplicator answers with $\beta(j)$, Duplicator loses since the resulting position does not respect the blocking pebble pair j. So assume otherwise that Duplicator answers with a different vertex w. The vertex w has to be the same color as $\alpha(j)$. However, there is exactly one vertex $w \equiv \beta(i)$ that has the same color as $\alpha(j)$, namely $\beta(j)$. Hence, $w \not\equiv \beta(i)$ but $\alpha(i) \equiv \alpha(j)$. This means that the resulting position is not a partial isomorphism and Spoiler wins.

   So consider the case where the position (α', β') in the game on $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$ does not induce a partial isomorphism (without blocking). For all $i \in [k]$ with $c(i) = c'(i) = \mathsf{regular}$, it holds

that $\alpha(i)$ and $\beta(i)$ have the same color—since Duplicator plays in a color-preserving manner—and thus $\alpha'(i) = \alpha(i)/_\equiv$ and $\beta'(i) = \beta(i)/_\equiv$ have the same color, too. If, for some regular pebble pairs i and j, we have $\alpha(i)/_\equiv = \alpha(j)/_\equiv$ but $\beta(i)/_\equiv \neq \beta(j)/_\equiv$, then $\alpha(i) \equiv \alpha(j)$ but $\beta(i) \not\equiv \beta(j)$ and Spoiler wins immediately. Suppose $\{\alpha(i)/_\equiv, \alpha(j)/_\equiv\}$ is an edge in $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ but $\{\beta(i)/_\equiv, \beta(j)/_\equiv\}$ is not an edge in $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$. On the one hand, there are vertices $u_i \equiv \alpha(i)$ and $u_j \equiv \alpha(j)$ such that $\{u_i, u_j\}$ is an edge in $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$. On the other hand, for every $v_i \equiv \beta(i)$ and $v_j \equiv \beta(j)$, the set $\{v_i, v_j\}$ is not an edge in $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$. Spoiler picks up a pebble pair different from the i-th and j-th, say the ℓ-th (such a pair exists because $k \geq 3$). Spoiler plays a regular move and places a pebble on u_i. Then Duplicator answers with some vertex v_i. If $v_i \not\equiv \beta(i)$, then Spoiler wins. Otherwise, Spoiler picks up the i-th pebble pair. Spoiler plays another regular round and places one pebble on u_j. Duplicator answers with some vertex v_j. If $v_j \not\equiv \beta(j)$, then Spoiler wins again. Otherwise, as already argued above, $\{v_i, v_j\}$ is not an edge, but $\{u_i, u_j\}$ is. Thus, Spoiler wins after 2 additional rounds.

The same arguments apply in the case that $\{\beta(i)/_\equiv, \beta(j)/_\equiv\}$ is an edge in $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$ but $\{\alpha(i)/_\equiv, \alpha(j)/_\equiv\}$ is not an edge in $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$.

We now prove by induction on $s \leq r$ that Spoiler has a winning strategy in the $(r+2)$-round game on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ such that the position reached in round s has an $(r-s)$-witness. Clearly, the initial position has an r-witness because Spoiler wins the r-round game $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$. Assume $s < r$ and that, by the induction hypothesis, the current position (α, β, c) of the game on $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ has an $(r-s)$-witness. If Spoiler removes a pair of pebbles, then we can update the position accordingly and the resulting position is still witnessed.

First, consider the case that Spoiler—according to their winning strategy on the compressed CFI graphs—plays a regular move in the $(r-s)$-witnessing position (α', β', c') on $\mathsf{CFI}(\mathcal{G}, f)/_\equiv$ and $\mathsf{CFI}(\mathcal{G}, g)/_\equiv$. Spoiler picks up the same pebble pair in the game on the precompressed CFI graphs $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ and plays a regular move, too. Assume Spoiler places the pebble on $u/_\equiv \in V_{\mathsf{CFI}(\mathcal{G}, f)/_\equiv}$ (the case for $u/_\equiv \in V_{\mathsf{CFI}(\mathcal{G}, g)/_\equiv}$ is analogous). Then, in the game on the precompressed CFI graphs, we let Spoiler place the pebble on u (which is not unique) and Duplicator answers with some vertex $v \in V_{(\mathsf{CFI}(\mathcal{G}, g), \equiv)}$. Since Spoiler had a winning strategy for $r-s$ rounds from position (α', β', c'), Spoiler has a winning strategy for $r-(s+1)$ rounds from position $(\alpha'[\ell \mapsto u/_\equiv], \beta'[\ell \mapsto v_\equiv], c'[\ell \mapsto \mathsf{regular}])$. Hence, on the compressed CFI graphs, the resulting position $(\alpha[\ell \mapsto u], \beta[\ell \mapsto v], c[\ell \mapsto \mathsf{regular}])$ has an $(r-(s+1))$-witness.

Second, consider the case that Spoiler plays a blocking move according to their strategy. We follow the strategy as in the previous case. If Spoiler places pebbles on $u/_\equiv \in V_{\mathsf{CFI}(\mathcal{G}, f)/_\equiv}$ and $v/_\equiv \in V_{\mathsf{CFI}(\mathcal{G}, f)/_\equiv}$ for the compressed CFI graphs, then we let Spoiler place pebbles on $u \in V_{(\mathsf{CFI}(\mathcal{G}, f), \equiv)}$ and $v \in V_{(\mathsf{CFI}(\mathcal{G}, f), \equiv)}$ for the precompressed CFI graphs. If Duplicator decides to place regular or blocking pebbles on u and v, then we consider the position in which Duplicator places regular or blocking, respectively, pebbles on $u/_\equiv$ and $v/_\equiv$. An induction on the number of blocking moves shows that the resulting position has an $(r-s-1)$-witness. □

Lemma 33 (Lemma 18 restated)**.** *Let $\equiv$ be a $\mathcal{G}$-compression and suppose $f, g : E_\mathcal{G} \to \mathbb{F}_2$ only twist a single edge e. If the robber, initially placed on the edge e, has a strategy for the first r rounds in the compressed and blocking k-Cops and Robber game on $\mathcal{G}$ and $\equiv$, then $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \simeq_\mathcal{B}^{k,r} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$.*

Proof. We show that Duplicator has a winning strategy in the r-round k-pebble game with blocking. Duplicator maintains a function $g' : E_\mathcal{G} \to \mathbb{F}_2$, an edge $e' \in E_\mathcal{G}$, and an isomorphism $\varphi : (\mathsf{CFI}(\mathcal{G}, g), \equiv) \to (\mathsf{CFI}(\mathcal{G}, g'), \equiv)$ such that after $s \leq r$ rounds in position (α, β, c) the following properties hold.

1. The isomorphism φ satisfies
   - $\varphi(\beta(i)) = \alpha(i)$ for every $i \in \mathsf{dom}(\alpha)$ such that $c(i) = \mathsf{regular}$, and
   - $\varphi(\beta(i)) \neq \alpha(i)$ for every $i \in \mathsf{dom}(\alpha)$ such that $c(i) = \mathsf{blocking}$.
2. Only the edge e' is twisted by f and g'.
3. At most one endpoint of e' is the origin of a vertex on which a pebble in placed by α. (Recall that the origin of a vertex $w = (u, \bar{a})$ in the (precompressed) CFI graph is the vertex u in the base graph $\mathcal{G}$).
4. The robber has a winning strategy in the $(r-s)$-round compressed and blocking k-Cops and Robber game starting in the following position:
   - The robber is placed on e'.
   - For every i with $c(i) = \mathsf{regular}$, a cop is placed on the $\equiv$-equivalence class of the origin of $\alpha(i)$.

- For every i with $c(i) = \mathsf{blocking}$ and for which $\alpha(i)$ and $\varphi(\beta(i))$ (and so $\beta(i)$) have the same origin x, the following roadblock is placed on $x/_{\equiv}$. Let $\alpha(i) = (x, \bar{a})$ and $\varphi(\beta(i)) = (x, \bar{b})$ and x be of degree d. We place the roadblock $\{i \in [d] \mid a_i \neq b_i\}$ on $x/_{\equiv}$, which by the definition of CFI graphs is a set of even size.

We first argue that maintaining the invariant for r rounds implies the statement of the lemma. By Conditions 2 and 3, at the end of round s for all $s \leq r$, the functions f and g' twist exactly one edge e' and at most one endpoint of e' is the origin of a vertex on which a pebble is placed. The edges $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g'), \equiv)$ only differ over the twisted connection. Over all other connections there is an edge between two vertices of $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ if and only if there is one in $(\mathsf{CFI}(\mathcal{G}, g'), \equiv)$. Since pebbles are placed on at most endpoint of the twisted connected, by applying Condition 1, we see that the pebbles induce a partial isomorphism with blocking between $(\mathsf{CFI}(\mathcal{G}, f), \equiv)$ and $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$. Therefore, Duplicator survives round s for all $s \leq r$ and so $(\mathsf{CFI}(\mathcal{G}, f), \equiv) \simeq^{k,r}_{\mathcal{B}} (\mathsf{CFI}(\mathcal{G}, g), \equiv)$.

We now show that Duplicator has a strategy to maintain the invariant. Because initially no pebbles are placed, the invariant clearly holds for $g' := g$ and $e' := e$. Assume, by the inductive hypothesis, that after $s < r$ many rounds the invariant holds and that in round $s + 1$ Spoiler picks up the ℓ-th pebble pair. If at the end of round s, the pair was placed on vertices of different colors (in which case $c(\ell) = \mathsf{blocking}$), we do nothing. Otherwise, the pair was placed on vertices of the same origin. In the Cops and Robber game, the Cops Player picks up the corresponding cop, if $c(\ell) = \mathsf{regular}$, or roadblock, if $c(\ell) = \mathsf{blocking}$, according to the invariant. Clearly, the invariant is maintained.

First consider the case where Spoiler makes a regular move and places a pebble on $u \in V_{(\mathsf{CFI}(\mathcal{G},f),\equiv)}$ (the case where $u \in V_{(\mathsf{CFI}(\mathcal{G},g),\equiv)}$ is analogous). Duplicator determines the destination for the pebble on $(\mathsf{CFI}(\mathcal{G}, g), \equiv)$ as follows. In the compressed and blocking k-Cops and Robber game, we suppose the Cops Player performs a cop move where they place a cop on the $\equiv$-class of the origin of u. Let T be the $\equiv$-compressible $\mathcal{G}$-twisting, with which the robber moves from e' to an edge e'' according to the robber's strategy. This compressible $\mathcal{G}$-twisting gives rise to an isomorphism $\psi\colon (\mathsf{CFI}(\mathcal{G}, g'), \equiv) \to (\mathsf{CFI}(\mathcal{G}, g''), \equiv)$ [25, Lemma 12] such that e' and e'' are precisely the edges twisted by g' and g''. Duplicator places the pebble on $\varphi^{-1}(\psi^{-1}(u))$.

We show that g'', e'', and $\psi \circ \varphi$ satisfy the invariant. Conditions 1 and 2 of the invariant hold because of the choice of T: We still have $\varphi(\beta(i)) = \alpha(i)$ for all $i \in \mathsf{dom}(\alpha)$ for which $c(i) = \mathsf{regular}$ because T avoids the cop on the origin x of $\beta(i)$ and thus the induced isomorphism ψ is the identity on the gadget of x [25, Lemma 13]. In particular, $\psi(\varphi(\beta(i))) = \varphi(\beta(i))$, which $\psi(\varphi(\beta(i))) = \alpha(i)$ by the inductive hypothesis. We also still have $\psi(\varphi(\beta(i))) \neq \alpha(i)$ for all $i \in \mathsf{dom}(\alpha)$ for which $c(i) = \mathsf{blocking}$ because T avoids the roadblock $\{(x, y_i) \mid a_i \neq b_i\}$, where $\alpha(i) = (x, \bar{a})$ and $\varphi(\beta(i)) = (x, \bar{b})$ and y_i is the i-th neighbor of x, on the origin x of $\beta(i)$. This implies that $\psi(\varphi(\beta(i))) \neq \alpha(i)$ by the definition of the induced isomorphism ψ (see again [25]). Condition 3 is satisfied because the robber is not caught in the game. For Condition 4, observe that since followed the strategy of the robber, the robber has a strategy for the remaining $(r - s - 1)$ rounds in the game. Moreover, since the roadblocks in the game are updated using the compressible $\mathcal{G}$-twisting T, they game are exactly the ones required in the game position by Condition 4 of the invariant. Thus Condition 4 holds. We finally update $g' \leftarrow g''$, $e' \leftarrow e''$, and $\varphi \leftarrow \psi \circ \varphi$.

It remains to consider the case where Spoiler makes a blocking move and places a pebble pair on $u \in V_{(\mathsf{CFI}(\mathcal{G},f),\equiv)}$ and $v \in V_{(\mathsf{CFI}(\mathcal{G},g),\equiv)}$. If u and v are not of the same color (and thus have different origins), Duplicator marks the pebble pair $\mathsf{blocking}$ and the invariant clearly holds. Otherwise, let the common origin of u and v be x and let $u = (x, \bar{a})$ and $\varphi(v) = (x, \bar{b})$. Let

$$N := \{\, i \in [d] \mid a_i \neq b_i \,\}.$$

If $N = \emptyset$, then we suppose the Cops Player makes a cop move and announce a cop for $x/_{\equiv}$. The robber moves with a $\equiv$-compressible $\mathcal{G}$-twisting T and we update e', g', and φ as in the case of a cop move. If T uses N, then we mark the pebble pair $\mathsf{regular}$. Otherwise, T avoids N and we mark the pebble pair $\mathsf{blocking}$. We show that the invariant is satisfied: If T uses N, then the isomorphism ψ, as defined in the regular move, is the identity on the gadget of x. Hence, we have $\varphi(v) = u$ also for the updated φ and Condition 1 is true. In the Cops and Robber game, a cop is placed on $x/_{\equiv}$ and since we followed the strategy of the Robber, Condition 4 holds. If T otherwise avoids N, ψ is not the identity for all vertices in the gadget for x. Hence, because we have $\bar{a} = \bar{b}$, we have $\psi(v) \neq u$ and thus $\varphi(\psi(v)) \neq u$ and Conditions 1 is true. Also, a roadblock is placed in the cops and robber game, which is exactly the one as in Condition 4.

If otherwise $N \neq \emptyset$, we suppose the Cops Player makes a blocking move and announces the roadblock N for $x/_{\equiv}$. Robber answers with $\equiv$-compressible twisting T with which the robber moves from the edge e' to the edge e''. We update e', g', and φ as in the case of a regular move. If T uses N, we will have that $u = \varphi(v)$ for the updated isomorphism φ. For the same reasons as in the regular move case, this ensures that Condition 1 is satisfied. Similarly, the remaining invariants hold for the same reasons as the regular move case.

If T avoids N, then $u \neq \varphi(v)$ for the updated isomorphism φ. The invariant holds for the same reason as before: Conditions 1 and 2 hold because of the choice of g' and e' and the properties of T, Condition 3 holds because the robber is not caught in the game, and Condition 4 holds because we followed the strategy of the robber (and again the roadblocks are updated after the robber moved). Now it is Spoiler's turn to make the next move, for which we proceed as before. An induction on the number of blocking moves in round $s+1$ shows that after the round, the invariant is satisfied. □

D Proofs and Material from Section 6

Discussion of the Generalized Construction. The original construction considers the case $t = 1$ and uses a cylindrical grid of length $J/2$ [25]. The generalized construction adds $f(k)$ columns to the begin and the end of the cylindrical grid but avoids the $f(k) < j, j' \leq J - f(k)$ condition [19]. Hence, the definition presented here can be seen as a variant of the one of [19], where we turn some non-singleton equivalence classes into singleton ones and cut off some columns at both ends. The former change only makes it easier for the robber and the latter change is not a restriction since there are still $f(k)$ columns containing only singleton classes. Moreover, the generalized construction puts the twists at the vertices and not the edges. This is unimportant since both variants of CFI graphs are isomorphic [18]. For simplicity, we use $f(k) = 4k$ instead of $f(k,t) = 2(k+t)$ so that J does not depend on t.

Formal Definitions of Periodic Paths and t-Critical Sets. We now provide the formal definitions of periodic paths and t-critical sets adapted from [19, 25], to which we refer for further explanations. We continue to work in the setting of Section 6. We may call $\equiv^t$-equivalence classes just $\equiv^t$-classes. We will also use sets of $\equiv^t$-classes W ti mean the set of all vertices contained in classes in W, so for the vertex set $\bigcup_{C \in W} C$, e.g., a $\mathcal{C}$-twisting T avoids a set of $\equiv^t$-classes W if T avoids $\bigcup_{C \in W} C$.

Definition 34 (Periodic Path)**.** *A path $(u_1, \ldots, u_m)$ in $\mathcal{C}$ is* ℓ-periodic *if u_1 is in the left and u_m is in right end, and for all $i < m$ and $v \in V_\mathcal{C}$ in the same row as u_i such that the distance of v and u_i is divisible by ℓ and u_i and v are not on the ends of $\mathcal{C}$, there is a $j < m$ such that $u_j = v$, and u_{i+1} and u_{j+1} are in the same row and their distance is divisible by ℓ.*

Intuitively, an ℓ-periodic path is composed out of a path of length ℓ, which is copied for all repetitions of ℓ columns. We continue with the formal definition of t-critical sets.

Definition 35 (Separating Sets)**.** *Let $t \in [k-1]$, $I \subseteq [k]$ and $q = \gcd\{f(k)p_i \cdot \ldots \cdot p_{i+t} \mid i \in I\}$. A set $W \subseteq V_\mathcal{C}$ is* (I,t)-separating *if there is no q-periodic path only using vertices of rows I such that the induced twisting avoids $\{(i,j) \mid (i,j') \in W, j' - j$ is divisible by $q\}$.*

Definition 36 (Virtual Cordon)**.** *A set $S \subseteq V_\mathcal{C}$ is a* vertical separator *if in $\mathcal{C} - S$ the left end is not connected to the right end of $\mathcal{C}$. For $t \in [k-1]$, a* t-virtual cordon *for a set $W \subseteq V_\mathcal{C}$ is a vertical separator S such that*

1. *for every row $i \in [k]$, the number of vertices in S lying in row i is at most the number of $\equiv^t$-classes, of which W contains vertices in row i, and*
2. *for every row $i \in [k]$ for which W contains only vertices of one $\equiv^t$-class c_i, we have $S \cap (\{i\} \times [J]) \subseteq c_i$, that is, S contains only vertices of c_i in row i.*

A minimal t*-virtual cordon is an inclusion-wise minimal t-virtual cordon. We call W* t-critical *if there exists a t-virtual cordon for W and W is (I,t)-separating for all $I \subseteq [k]$ of size at most $t+1$.*

This means, for rows in which W contains vertices of exactly one $\equiv^t$-class C, a t-virtual cordon S for W contains exactly one vertex of the class C.

Cops do not Benefit From Roadblocks. We now give a detailed proof of Theorem 25, which follows by showing a lower bound on the round number of the compressed and blocking Cops and Robbers game played on $\mathcal{C}$ and $\equiv^t$. To do this we first need two auxiliary lemmas. The first says that if there are not too many cops placed, then the robber can always move from one end of the grid to the other.

Lemma 37 (Lemma 23 restated)**.** *Let $t \in [k-1]$ and $c \leq \frac{2}{5}k - 1$ be integers. Consider the compressed and blocking $(k+c)$-Cops and Robber game on $\mathcal{C}$ and $\equiv^t$ and assume that cops are placed in at most c rows. Then there is a t-end-to-end twisting that avoids all cop-occupied $\equiv^t$-equivalence classes and avoids all roadblocks.*

Proof. We can assume that cops are placed in exactly c rows. Otherwise, we convert roadblocks into cops until cops are placed in exactly c rows. As already argued, converting roadblocks to cops only makes it harder to find the desired t-end-to-end twisting (or to move the robber in general). If this is not possible, then there is a row not containing cops and roadblocks, which means that the path straight through this row is the desired t-end-to-end twisting.

Let $I \subseteq [k]$ be the set of rows in which no cop is placed. We call these rows *cop-free*. Clearly, there are $(k-c)$ cop-free rows. A cop-free row $i \in I$ is *lonely* if $i-1 \notin I$ and $i+1 \notin I$ (recall that we use addition on rows such that the $(k+1)$-th row is the 1st one). A cop-free row is *socialized* if it is not lonely. Let ℓ be the number of lonely rows. Hence, there are $s := k - c - \ell$ socialized rows.

A *horizontal* roadblock is a roadblock that blocks the use of exactly the two horizontal edges of a vertex or $\equiv^t$-class, the one to the left and the one to the right. A roadblock *belongs* to a row $i \in I$ if

(a) it is a horizontal roadblock contained in row i, or

(b) it is not contained in row i but blocks an edge incident to a vertex in row i and a horizontal edge in another row (and hence a roadblock can belong to at most one row).

If there is a cop-free row that does not contain a horizontal roadblock, then the path straight through this row is a desired t-end-to-end twisting. Hence, for every cop-free row i, there is a horizontal roadblock that belongs to row i unless the desired t-end-to-end twisting exists.

Let $i \in I$ be a socialized row and assume that $i+1 \in I$ (the case that $i-1 \in I$ is analogous by considering $i-1$ instead of i). We show that, unless the desired t-end-to-end twisting exists, there have to be two non-horizontal roadblocks in addition to the horizontal roadblocks from before, one that belongs to row i blocking an edge incident to a vertex in row $i+1$ and one that belongs to row $i+1$ and blocks an edge incident to a vertex in row i. Consider the two paths:

$$\pi_1 = \big((i,0),(i,1),(i+1,1),(i+1,2),(i,2)\big)$$
$$\pi_2 = \big((i+1,0),(i+1,1),(i,1),(i,2),(i+1,2)\big),$$

Each of them induces a 2-periodic path by copying π_1 and π_2, respectively, to every second column because π_1 starts and ends in row i and π_2 starts and ends in row $i+1$ (the 2-periodic paths by are drawn in Figure 2). These 2-periodic paths induce t-end-to-end twistings by Lemma 21, since 2 is a divisor of $f(k)$. Clearly, the induced twistings avoid every possible horizontal roadblock. Moreover, for every vertex v, the incident edges of v used by the two induced twistings differ from one another. This means that a roadblock that is not avoided by π_1 is avoided by π_2 and vice versa. Hence, there have to be at least two additional non-horizontal roadblocks in rows i and $i+1$ unless there is a desired t-end-to-end twisting. Every roadblock which blocks the path π_1 or π_2 belongs to row i or $i+1$, respectively. Hence, for each socialized row there are at least two non-horizontal roadblocks belonging to it unless an t-end-to-end twisting exists.

So suppose for a contradiction that no t-end-to-end twisting exists. We now count the number of roadblocks. For every cop-free row i, we have shown that up to three additional roadblocks belonging to row i have to exist: one horizontal roadblock and two non-horizontal roadblocks. The horizontal roadblock always has to exist to prevent the path straight through row i from being a t-end-to-end twisting. Whether the non-horizontal roadblocks have to exist depends on whether there are cops in rows $i-1$ and $i+1$: we require a non-horizontal one blocking an edge incident to row $i+1$ (if this row is also cop-free) and a non-horizontal one blocking an edge incident to row $i-1$ (if this row is also cop-free). A row containing a cop can reduces the number of roadblocks needed in some other row by one for at most two cop-free rows. If row i contains a cop, then the required number of non-horizontal roadblock in rows $i-1$ and $i+1$ is reduced by one (if they are cop-free). Since $(k-c)$ rows do not contain a cop, there have to be at least $3(k-c)-2c$ roadblocks. Since c rows contain a cop and as we consider the compressed and blocking $(c+k)$-Cops and Robber game, there are at most k roadblocks. This implies

$$3(k-c)-2c \leq k \qquad \text{and thus}$$
$$\frac{2}{5}k \leq c$$

This is a contradiction to the assumption that $c \leq \frac{2}{5}k - 1$. Hence, the desired t-end-to-end twisting must exist. □

Lemma 38. *Let $t \in [k-1]$ and $W_1, \ldots, W_m$ be sets of at most $(k+t)$ many $\equiv^t$-equivalence classes such that $|W_i \cup W_{i+1}| \leq k+t$ for all $i < m$. Let $I \subseteq [k]$ be a set of rows, of which all W_i contain exactly one and the same class. Also assume that all W_i are t-critical. Then all minimal t-semi-separators for each W_i coincide on the rows in I. In particular, if I is nonempty, the* diameter *of all vertex minimal t-semi-separators together*

is at most $2(k+t)$. This means that there is a j such that all these t-semi-separators are contained within column $j, \dots, j+2(k+t)-1$.

Proof. We prove, for every $j < m$, that the statement holds for the sets $W_1, \dots, W_j$ by induction on j. For $j = 1$, the statement obviously holds. For the inductive step, assume that all minimal t-semi-separators for each W_i with $i \leq j$ agree on the rows in I. Now consider the set $W = W_j \cup W_{j+1}$. By assumption, W has size at most $k+t$. By [19, Proposition 4.9], all minimal t-semi-separators for W agree on the rows in I. Combined with the inductive hypothesis, this implies that all minimal t-semi-separators for each W_i with $i \leq j+1$ agree on the rows in I, since every minimal t-semi-separator for W_{i+1} is also one for W [19].

Now assume that I is nonempty. So all minimal t-semi-separators for each W_i with $i \in [m]$ share at least one vertex. Since the W_i are of size at most $k+t$, every t-semi-separator for them has size at most $k+t$. Hence, their individual diameter is at most $k+t$ [19]. Since all these t-semi-separators share a vertex, their diameter together is at most $2(k+t)$. □

We now use the two preceding lemmas to prove our round number lower bound.

Lemma 39 (Lemma 24 restated)**.** *Let $1 \leq t \leq \frac{2}{5}k - 1$ be an integer. Then the robber, initially placed on the left or right end of $\mathcal{C}$, has a strategy for the first $\Omega(J)$ rounds in the compressed and blocking $(k+t)$-Cops and Robber game on $\mathcal{C}$ and $\equiv^t$.*

Proof. We use letters P and Q for positions in the compressed and blocking $(k+t)$-Cops and Robber game. For such a position P, we write $\hat{P}$ for the position in which every roadblock is turned into a cop, that is, if there is a roadblock for a vertex u in P, then we put a cop on u in $\hat{P}$. Existing cops are copied. We call such a position $\hat{P}$ t-critical, if the set of cop-occupied classes in $\hat{P}$ is t-critical. Similarly, a t-virtual cordon for $\hat{P}$ is a t-virtual cordon for the set of cop-occupied classes in $\hat{P}$.

As already argued for Lemma 23, replacing roadblocks with cops only makes the game harder for the robber. So, if the robber has a strategy for the next r rounds from position $\hat{P}$, then the robber also has such a strategy from position P. Moreover, $\hat{P}$ can also be seen as a position in the non-blocking game.

Claim 40. *Let P be a position in the game such that $\hat{P}$ is t-critical and the robber has distance at least r to every minimal t-virtual cordon for $\hat{P}$. Then the robber has a strategy in the compressed and blocking $(k+t)$-Cops and Robber game in position P for the next round, resulting in a position Q, such that the robber has distance at least $r - O(k)$ to every minimal t-virtual cordon for $\hat{Q}$.*

Proof of claim. We prove the statement by induction on the number of moves in the current round as follows: Let $P = Q_0, Q_1, \dots, Q_m$ be the sequence of positions during the first m moves of the current round. Also let $Q_0^-, \dots, Q_{m-1}^-$ be the intermediate positions in which a cop or roadblock was picked up. This means, that the game changes from position Q_i to Q_i^- to Q_{i+1} for every $0 \leq i < m$. We show that the robber has a strategy such that all positions do not contain empty roadblocks and either

1. all the positions $\hat{Q}_i^-$ are t-critical and every minimal t-virtual cordon for $\hat{Q}_m$ has distance at least $r - O(k)$ to the robber, or
2. for the minimal $\ell \in [m]$ such that $\hat{Q}_i^-$ is t-critical for all $\ell \leq i \leq m-1$, the robber has distance $\Omega(J)$ to the minimal t-virtual cordons for $\hat{Q}_i$ for each $\ell \leq i \leq m$.

For the base case $m = 0$, the inductive hypothesis clearly holds. So assume that after m necessarily blocking moves with roadblock responses (otherwise the round already ended) the game is in position Q_m and the inductive hypothesis is satisfied for m. Now a cop or roadblock is picked up resulting in position Q_m^-. We make the following case distinction:

(a) If $\hat{Q}_m^-$ is not t-critical, then we show that if the next position $\hat{Q}_{m+1}$ is t-critical, then the distance of the robber to every t-virtual cordon in $\hat{Q}_{m+1}$ is $\Omega(J)$.

This is straightforward to see if the Cops Player makes a cop move. We just follow the strategy in the non-blocking game in position $\hat{Q}_m^-$ to ensure this distance [19, Proof of Theorem 4.1]. The robber moves to the end with larger distance to all the t-virtual cordons. Since different minimal t-virtual cordons can have distance at most $O(k)$ [19], one end has distance at least $J/2 - O(k)$ to these t-virtual cordons. Otherwise, the Cops Player makes a blocking move. Let N be the announced roadblock on a $\equiv^t$-equivalence class C. Moreover, imagine that in the non-blocking game a cop is announced for C; then the robber responds with some appropriate $\equiv^t$-compressible twisting—which we denote by T—according to the robber's strategy. Call the resulting position in the non-blocking game Q, in which the robber again has distance $O(J)$ to all t-virtual cordons. The robber moves using T also in the blocking game. If T uses N a cop is placed on C and the resulting position is Q_{m+1}. But observe that $\hat{Q}_{m+1}$ is identical to Q. Otherwise, T avoids N. Then a roadblock is placed and again, the resulting position Q_{m+1} is such that $\hat{Q}_{m+1}$ is identical to Q.

(b) Now assume that $\hat{Q}_m^-$ is t-critical and, moreover, that in position Q_m^- there are cops in at most $\frac{2}{5}k-1$ rows. Then there is a $\equiv^t$-compressible end-to-end twisting T with which the robber can move to the other end of $\mathcal{C}$ by Lemma 23. If the Cops Player makes a cop move, the robber can use T to move to the end with larger distance to all minimal t-virtual cordons (which might appear after the cop is placed). The robber may need to avoid cops placed in the ends. This is always possible because all minimal t-virtual cordons are not in the end where the robber is located. Note that there are only singleton equivalence classes in both ends so inside the ends the robber can move easily. From now we silently assume that, after possibly moving sides, the robber also avoid these non-dangerous cops inside ends locally. This means that all twisting we use to move the robber get modified for this in-end avoiding of cops.

Otherwise the Cops Player makes a blocking move. In this case the robber first checks whether the robber they should switch ends, i.e. whether the robber is at the end with larger distance to all minimal t-virtual cordons. If switching is not required, define T' to be the trivial twisting with which the robber does not move at all. If switching ends is required, let $T' := T$. The robber moves using twisting T'.

(c) Otherwise, $\hat{Q}_m^-$ is t-critical and there are at least $\frac{2}{5}k$ rows in which cops are placed in Q_m^-. In this case the robber stays in the end where th robber is currently located. If the Cops Player makes a cop move, then the robber does not move (apart from possibly to avoid in-end cops). If the Cops Player announces a roadblock N on a $\equiv^t$-equivalence class C, then the robber similarly only moves to avoid in-end cops and roadblocks.

If the resulting position $\hat{Q}_{m+1}$ is not t-critical, we are done. Otherwise, let $\ell \in \{0,\dots,m\}$ be the minimal number, such that $\hat{Q}_i^-$ is t-critical for all $\ell \leq i \leq m$. Let W be a minimal t-virtual cordon for $\hat{Q}_\ell^-$. Every other minimal t-virtual cordon for $\hat{Q}_\ell^-$ has distance at most $O(k)$ to W [19].

Let W' be a minimal t-virtual cordon for $\hat{Q}_m^-$. Let I be the set of rows, in which there is exactly one cop in both Q_ℓ^- and in Q_{m+1}^-. We claim that $|I| \geq 1$. To see this note that no new cops are placed when roadblocks are placed, that there are at most $k+(\frac{2}{5}k-1)$ cops in Q_ℓ^-, and that there are cops in at least $\frac{2}{5}k$ rows in Q_{m+1}^-.

The positions $\hat{Q}_\ell^-,\dots,\hat{Q}_m^-,\hat{Q}_{m+1}$ satisfy the prerequisites of Lemma 38: all of them are t-critical and in each step exactly one cop is moved (and for $\hat{Q}_m^-$ and $\hat{Q}_{m+1}$ one cop is added). Hence, W' and W together have diameter at most $2(k+t)=O(k)$. If $\ell=0$, then the robber has distance at least $r-O(k)$ to every minimal t-virtual cordon for $\hat{Q}_{m+1}$. Otherwise, the robber has distance at least $\Omega(J)-O(k)=\Omega(J)$ to every minimal t-virtual cordon for $\hat{Q}_{m+1}$. □

Claim 40 shows that in each round where the position is t-critical, independent of the number of roadblock responses, the cops can move a minimal virtual cordon by at most $O(k)$ many columns towards the robber without providing the robber the opportunity to switch between the ends. If the cops allow the robber to switch ends, we essentially can start the game from the beginning again because the robber can enforce a distance of $\Omega(J)$ to all minimal t-virtual cordons. This means that we can follow the strategy for the robber used in the non-blocking game as outlined before [19], where after each round we convert all roadblocks to cops. If the position is not critical, it does not matter where the robber is. If the position becomes critical, the robber moves to the end with larger distance to the minimal virtual cordons. If the position is critical, then by Claim 40, this distance of the robber to the nearest t-virtual cordon decreases by only $O(k)$ per round. Hence this strategy can be maintained for $\Omega(J/O(k))$ rounds. Together with the observation that a lower bound for $\hat{P}$ in the blocking game implies a lower bound for P in the blocking game, this implies a $\Omega(J/O(k))=\Omega(J)$ lower bound for the robber in the blocking game since k is considered a constant. □

Theorem 41 (Theorem 25 restated)**.** *For all integers $k \geq 3$, $1 \leq t \leq \frac{2}{5}k-1$, and $n \in \mathbb{N}$, there are two colored graphs $\mathcal{G}$ and $\mathcal{H}$ of order $\Theta(n)$ and color class size 8 such that*

1. *$\mathcal{G} \not\simeq^{k+1} \mathcal{H}$, that is, Spoiler wins the $(k+1)$-pebble game on $\mathcal{G},\mathcal{H}$, and*
2. *$\mathcal{G} \simeq_{\mathcal{B}}^{k+t,\Omega(n^{k/(t+1)})} \mathcal{H}$, that is, Duplicator has a strategy for the first $\Omega(n^{k/(t+1)})$ rounds in the $(k+t)$-pebble game with blocking on $\mathcal{G},\mathcal{H}$.*

Proof. Let $k \geq 3$, $1 \leq t \leq \frac{2}{5}k-1$, and $n \in \mathbb{N}$. If n is sufficiently large, for $w = \lceil \sqrt[t+1]{n} \rceil$ the required pairwise coprime numbers $p_1,\dots,p_k$ exist [25]. Let $J = f(k)p_1 \cdot \ldots \cdot p_k$ and $\mathcal{C}$ be the $k \times J$ cylindrical grid. Also, let $\equiv^t$ be the compression defined earlier with respect to $p_1,\dots,p_k$. By Lemma 24, the robber has a strategy for the first $\Omega(J)=\Omega(w^k)=\Omega(n^{k/(t+1)})$ rounds in the compressed and blocking $(k+t)$-Cops and Robber game on $\mathcal{C}$ and $\equiv^t$, when the robber is placed on the left or right end.

Now let e be an edge in the left end, $f,g : E_{\mathcal{C}} \to \mathbb{F}_2$ only twist e, and set $\mathcal{G} := \mathsf{CFI}(\mathcal{C},f)/_{\equiv^t}$ and $\mathcal{H} := \mathsf{CFI}(\mathcal{C},f)/_{\equiv^t}$. By Lemmas 14 and 18, Duplicator has a strategy for the first $\Omega(n^{k/(t+1)})$ rounds of the

$(k+t)$-pebble game with blocking on $\mathcal{G}$ and $\mathcal{H}$. It is known that Spoiler wins the $(k+1)$-pebble game on CFI graphs over cylindrical grids and thus over the compressed ones by Lemma 14. By Lemma 19, the graphs $\mathcal{G}$ and $\mathcal{H}$ have size $\Theta(n)$ and are of color class size 8. □

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